

Fourier Decay of Rvachev's Up-Function

A guided and rigorous account of exact products, decay spectra, fluctuations, Cesàro gauges, peaks, and valleys

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Abstract

Let

$$\Phi(x) = \widehat{\text{up}}(x) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^h}\right), \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

The transform is entire and decreases faster than every inverse power, but it has zeros at all nonzero integers and its nonzero values fluctuate on many different scales. Therefore there is no single unqualified “decay rate.” The exact dyadic identity

$$|\Phi(2^n y)| = |\Phi(y)| \pi^{-n} y^{-n} 2^{-n(n+1)/2} \prod_{j=1}^n |\sin(\pi 2^j y)|$$

separates a deterministic log-normal factor from a lacunary dynamical product. This separation yields a spectrum of powers, according to whether size means a typical value, an arithmetic mean, a root-mean-square value, or a uniform envelope. With

$$E_{\kappa}(x) = x^{-\kappa} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right),$$

the four principal exponents are

$$\begin{aligned} \kappa_0 &= 3.151496129472318798\dots, & \kappa_1 &= 2.748070014871332674\dots, \\ \kappa_2 &= 2.651496129472318798\dots, & \kappa_{\infty} &= 2.359014879111740707\dots \end{aligned}$$

The first, third, and fourth have elementary closed forms; κ_1 is determined by the Perron root of an explicit weighted doubling operator and comes with a rigorous rational enclosure.

The article gives detailed proofs of the product identities, zero multiplicities, unique lobe maxima, exact dyadic renormalization, geometric-mean and almost-everywhere laws, exact logarithmic variance, the martingale decomposition, the root-mean-square identity, the sharp Gelfond envelope, Lambert- W boundary-layer asymptotics for dyadic-annulus maxima, and explicit dyadic valley laws. It also treats the arithmetic-mean/Cesàro problem: the natural fixed gauge works shell-by-shell and at dyadic endpoints, a singular phase profile obstructs an unrestricted limit for every bounded-distortion correction, and a deliberately shell-adaptive positive analytic decreasing gauge does achieve the unrestricted limit. Finally, the arguments are extended to products with an arbitrary integer geometric ratio. Imported functional-analytic and probabilistic theorems are stated with their hypotheses, and numerical evidence is kept visibly separate from exact proof.

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1 What is being measured?

Rvachev's up-function is an even, nonnegative, compactly supported C^∞ function on $[-1, 1]$ with total mass one. Under the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i t \xi} dt, \quad (1.1)$$

its transform is Φ above. The central question is deceptively simple: *how fast does $\Phi(x)$ tend to zero?*

The first answer is that the question needs a qualifier. Four facts force this.

1. $\Phi(m) = 0$ for every nonzero integer m , so no positive function g can satisfy $|\Phi(x)| \sim g(x)$ on the entire real ray.
2. Between consecutive integers there is exactly one maximum of $|\Phi|$, but those maxima themselves have exceptional subsequences of very different sizes.
3. On a dyadic shell, the geometric mean, arithmetic mean, quadratic mean, and supremum of the oscillatory numerator grow at different exponential rates.
4. Even after the correct arithmetic-mean normalization is chosen, an incomplete final dyadic shell remembers its position within the shell.

The universal part of the answer is the function

$$E_\kappa(x) := x^{-\kappa} \exp\left(-\frac{(\log x)^2}{2a}\right), \quad a := \log 2, \quad x > 0. \quad (1.2)$$

The coefficient $1/(2 \log 2)$ is forced by the triangular product of dyadic denominators. The power κ , however, depends on how the remaining sine product is sampled.

Reading guide

The article has a short logical spine.

1. [Section 2](#) proves the exact product and zero structure.
2. [Section 3](#) derives the master dyadic identity and explains the log-square term.
3. [Sections 4, 5, 7 and 8](#) obtain the four main exponents.
4. [Section 6](#) studies typical fluctuations beyond the leading exponent.
5. [Sections 9 to 12](#) answer the additive and logarithmic averaging questions.
6. [Sections 13, 15 and 16](#) describe large peaks, moving annular maxima, and deep exceptional valleys.
7. [Section 17](#) gives the integer-ratio generalization.

Readers interested only in the principal decay laws may read through [Section 8](#), then jump to [Section 13](#) and the summary in [Section 18](#).

1.1 The four principal scales

For $q \in (0, \infty)$, let the normalized L^q size on the dyadic shell $D_n = [2^{n-1}, 2^n]$ be

$$\mathcal{M}_{q,n}(H) = \left(\frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} |H(x)|^q dx \right)^{1/q}. \quad (1.3)$$

The endpoint conventions are the geometric mean $q = 0$ and the supremum $q = \infty$. The main scale dictionary is as follows.

Notion of size	numerator rate	exponent in E_κ	interpretation
Typical / geometric mean	$1/2$	$\kappa_0 = \frac{3}{2} + \frac{\log \pi}{\log 2}$	Exact shell geometric mean; also the logarithmic rate on almost every fixed dyadic ray.
Arithmetic mean	$\rho_1 = 0.6613226020 \dots$	$\kappa_1 = \frac{\log \pi + \frac{1}{2} \log 2 - \log \rho_1}{\log 2}$	Natural normalization for shell L^1 means and ordinary Cesàro averages.
Root mean square	$2^{-1/2}$	$\kappa_2 = \frac{\log(2\pi)}{\log 2}$	Exact finite-level Parseval identity.
Uniform envelope	$\sqrt{3}/2$	$\kappa_\infty = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2}$	Sharp global pointwise upper bound, attained up to a fixed constant on a period-two ray.

Numerically,

$$\begin{aligned} \kappa_\infty = 2.3590148791 \dots < \kappa_2 = 2.6514961295 \dots \\ < \kappa_1 = 2.7480700149 \dots < \kappa_0 = 3.1514961295 \dots \end{aligned} \quad (1.4)$$

Larger q gives more weight to rare large values, so it produces slower decay and hence a smaller exponent.

Common pitfall

The power $\log_2(2\pi)$ is neither an approximation nor a universally correct answer. It is exactly the *root-mean-square* exponent κ_2 . The other notions of size differ already by a nonzero power of x ; multiplying by $(\log x)^p$ cannot reconcile them.

1.2 A compact statement of the final answer

Theorem 1.1 (Decay dictionary). *Let E_κ be defined by (1.2).*

1. For Lebesgue-almost every fixed $y \in [1/2, 1]$,

$$|\Phi(2^n y)| = E_{\kappa_0}(2^n y) e^{o(n)}.$$

The centered logarithm has asymptotic variance $\pi^2/4$ per dyadic level and obeys the central limit theorem and the law of the iterated logarithm stated in Theorem 6.4.

2. For every fixed finite $q > 0$, there is $A_q \in (0, \infty)$ such that

$$\mathcal{M}_{q,n} \left(\frac{\Phi}{E_{\kappa_q}} \right) \longrightarrow A_q,$$

where κ_q is determined by the leading eigenvalue of the weighted transfer operator in Section 7.

3. There is $C < \infty$ such that

$$|\Phi(x)| \leq C E_{\kappa_\infty}(x), \quad x \geq 1,$$

and this exponent is sharp on $x = 2^n(2/3)$.

4. The maximum on $[2^k, 2^{k+1}]$, when compared with the envelope at the shell's left endpoint, has the additional Lambert–W suppression of Theorem 15.4.
5. The natural arithmetic-mean gauge $A_1 E_{\kappa_1}$ gives bounded ordinary Cesàro means and convergence at dyadic endpoints. At arbitrary endpoints it leaves a nonconstant phase profile. No bounded-distortion shell correction can remove that profile, but the adaptive analytic gauge of Theorem 11.4 does give the unrestricted limit.

6. The fixed natural scale has an unrestricted limit under logarithmic rather than additive averaging.

The remainder of the article proves these statements in an order designed to make each new object inevitable rather than mysterious.

2 The entire product, its zeros, and its lobes

This section develops everything that can be obtained from elementary complex analysis and real-variable calculus before any ergodic or spectral machinery enters.

2.1 Normal convergence and three product representations

Proposition 2.1 (Entire convergence). *The infinite product*

$$\Phi(z) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right) \quad (2.1)$$

converges locally uniformly on $\mathbb{C}$. It defines an even entire function, real-valued on $\mathbb{R}$, with $\Phi(0) = 1$.

Proof. Fix a compact set $K \subset \mathbb{C}$, and choose R with $|z| \leq R$ on K . The Taylor expansion at the origin is

$$\operatorname{sinc} w = 1 - \frac{w^2}{6} + O(w^4).$$

Hence there are h_0 and C_R such that, for $h \geq h_0$ and $z \in K$,

$$\left| \operatorname{sinc}(\pi z/2^h) - 1 \right| \leq C_R 4^{-h}.$$

The majorant $\sum_{h \geq h_0} C_R 4^{-h}$ converges. The standard normal-convergence criterion for infinite products therefore gives locally uniform convergence of the tail, except possibly at zeros of a finite initial factor; multiplying by that finite prefix still gives an entire limit. Every factor is even, is real on the real axis, and equals one at zero. These properties pass to the locally uniform limit. $\square$

There are two other representations that expose different structures.

Theorem 2.2 (Cosine and zero-divisor products). *For every $z \in \mathbb{C}$,*

$$\Phi(z) = \prod_{j=1}^{\infty} \cos^j\left(\frac{\pi z}{2^j}\right) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}. \quad (2.2)$$

Both products converge locally uniformly.

Proof. The elementary double-angle identity gives, for every integer $N \geq 1$,

$$\sin t = 2^N \sin(t/2^N) \prod_{j=1}^N \cos(t/2^j).$$

After division by t and passage to the limit $N \rightarrow \infty$, using $2^N \sin(t/2^N)/t \rightarrow 1$, we obtain Viète's product

$$\operatorname{sinc} t = \prod_{j=1}^{\infty} \cos(t/2^j).$$

Insert $t = \pi z/2^h$ into (2.1). The factor $\cos(\pi z/2^j)$ occurs once for every pair (h, r) with $h \geq 0$, $r \geq 1$, and $h + r = j$; there are exactly j such pairs. Local absolute convergence of the logarithms away

from zeros, followed by analytic continuation across the discrete zero set, justifies the regrouping and proves the first equality.

For the second equality use Euler's sine product

$$\operatorname{sinc}(\pi w) = \prod_{\ell=1}^{\infty} \left(1 - \frac{w^2}{\ell^2}\right).$$

At level h , the zero factors are $1 - z^2/(2^h \ell)^2$. A fixed positive integer m occurs once for each representation $m = 2^h \ell$, equivalently for each $0 \leq h \leq \nu_2(m)$; its multiplicity is therefore $1 + \nu_2(m)$. On any compact set,

$$\sum_{h \geq 0} \sum_{\ell \geq 1} \frac{|z|^2}{4^h \ell^2} < \infty,$$

so the double product is normally convergent and may be rearranged. This proves the second equality. $\square$

2.2 Scaling, zeros, and local zero coefficients

The product immediately implies the one-step relations

$$\Phi(2x) = \operatorname{sinc}(2\pi x)\Phi(x), \quad \Phi(x) = \operatorname{sinc}(\pi x)\Phi(x/2). \quad (2.3)$$

The first is obtained by separating the $h = 0$ factor from $\Phi(2x)$; the second is the same identity with x replaced by $x/2$.

Proposition 2.3 (Integer zeros and their exact orders). *The nonzero zeros of Φ are precisely the nonzero integers. At $m \in \mathbb{Z} \setminus \{0\}$, the multiplicity is*

$$d(m) = 1 + \nu_2(|m|). \quad (2.4)$$

If $b \geq 1$ is an integer, $d = 1 + \nu_2(b)$, and $z \uparrow b$, then

$$|\Phi(z)| = b^{-d} \left| \Phi(b/2^d) \right| (b - z)^d (1 + O(b - z)). \quad (2.5)$$

Proof. The last product in (2.2) shows both the zero set and the multiplicity. For the coefficient, return to the sinc product. Exactly the factors with $0 \leq h \leq d - 1$ vanish at $z = b$. Each zero is simple, and direct differentiation gives

$$\left| \frac{d}{dz} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right) \right|_{z=b} = \frac{1}{b}.$$

Thus the product of the d vanishing factors contributes $b^{-d}(b - z)^d(1 + O(b - z))$. The remaining factors are nonzero at b and their value is

$$\prod_{h=d}^{\infty} \left| \operatorname{sinc}\left(\frac{\pi b}{2^h}\right) \right| = \left| \Phi(b/2^d) \right|.$$

Multiplying the two parts proves (2.5). $\square$

2.3 Exactly one extremum in every unit interval

The zero-divisor product does more than locate zeros: its logarithm is strictly concave between consecutive zeros.

Theorem 2.4 (Strict logarithmic concavity and unique lobe maxima). *The function Φ is positive and strictly decreasing on $(0, 1)$. For every integer $m \geq 1$, the function $|\Phi|$ has exactly one critical point $\xi_m \in (m, m+1)$, and this point is the strict maximum on that interval. Moreover,*

$$\operatorname{sgn} \Phi(x) = (-1)^{s_2(m)}, \quad m < x < m+1, \quad (2.6)$$

where $s_2(m)$ is the number of ones in the binary expansion of m .

Proof. Fix a compact subinterval avoiding the integers. Since $\sum_{k \geq 1} (1 + \nu_2(k))/k^2 < \infty$, the logarithm of the last product in (2.2) may be differentiated term by term. We obtain

$$\frac{d}{dx} \log |\Phi(x)| = -2x \sum_{k=1}^{\infty} \frac{1 + \nu_2(k)}{k^2 - x^2}, \quad (2.7)$$

$$\frac{d^2}{dx^2} \log |\Phi(x)| = -2 \sum_{k=1}^{\infty} (1 + \nu_2(k)) \frac{k^2 + x^2}{(k^2 - x^2)^2} < 0. \quad (2.8)$$

For $0 < x < 1$, every denominator in (2.7) is positive, hence the logarithmic derivative is negative. This proves strict decrease on the central lobe.

Now fix $m \geq 1$. Equation (2.8) says that the logarithmic derivative is strictly decreasing on $(m, m+1)$. The term $k = m$ forces it to tend to $+\infty$ as $x \downarrow m$, while the term $k = m+1$ forces it to tend to $-\infty$ as $x \uparrow m+1$. The intermediate value theorem and strict monotonicity give exactly one zero. Strict log-concavity makes that zero the unique maximum of the modulus.

It remains to compute the sign. Starting from the positive central lobe, crossing the zero at k changes the sign $1 + \nu_2(k)$ times. Therefore the sign on $(m, m+1)$ is

$$(-1)^{\sum_{k=1}^m (1 + \nu_2(k))} = (-1)^{m + \nu_2(m!)}.$$

Legendre's formula in base two is $\nu_2(m!) = m - s_2(m)$. Hence the exponent is $2m - s_2(m)$, which has the same parity as $s_2(m)$. This proves (2.6). $\square$

What the formula says

The graph is not irregular in the small: every interval $(m, m+1)$ contains one clean, strictly log-concave lobe. The irregularity is in the *heights and locations* of those lobes as m ranges through integers with different binary structure.

2.4 Reciprocal Gamma and corrected q -Pochhammer tails

These factorizations are useful for local evaluation and for separating finitely many nearby zeros from a rapidly convergent analytic tail.

Theorem 2.5 (Gamma product). *For every $z \in \mathbb{C}$,*

$$\Phi(z) = \prod_{n=0}^{\infty} \frac{1}{\Gamma(1 + z/2^n) \Gamma(1 - z/2^n)}. \quad (2.9)$$

Proof. Euler's reflection formula and $\Gamma(1+w) = w\Gamma(w)$ imply

$$\Gamma(1+w)\Gamma(1-w) = \frac{\pi w}{\sin \pi w},$$

so its reciprocal is $\operatorname{sinc}(\pi w)$. Substitute $w = z/2^n$ and multiply over $n \geq 0$. Local uniform convergence follows from the same $1 + O(4^{-n})$ estimate as in Theorem 2.1. $\square$

For $|q| < 1$, write

$$(u; q)_\infty = \prod_{j=0}^{\infty} (1 - uq^j), \quad c_k = \frac{\zeta(2k) - 1}{k} > 0.$$

Theorem 2.6 (Zero-extracted Pochhammer tail). *For an integer $m \geq 0$, define*

$$R_m(z) = \prod_{n=m}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right).$$

If $|z| < 2^{m+1}$, then

$$R_m(z) = (z^2/4^m; 1/4)_\infty \exp\left[-\sum_{k=1}^{\infty} \frac{c_k}{1-4^{-k}} \left(\frac{z}{2^m}\right)^{2k}\right]. \quad (2.10)$$

Consequently,

$$\Phi(z) = \left(\prod_{n=0}^{m-1} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right)\right) R_m(z), \quad (2.11)$$

with the empty prefix interpreted as one.

Proof. Euler's sine product gives, for $|w| < 2$,

$$\begin{aligned} \operatorname{sinc}(\pi w) &= (1 - w^2) \prod_{\ell=2}^{\infty} \left(1 - \frac{w^2}{\ell^2}\right), \\ \log \frac{\operatorname{sinc}(\pi w)}{1 - w^2} &= \sum_{\ell=2}^{\infty} \log \left(1 - \frac{w^2}{\ell^2}\right). \end{aligned}$$

The double series obtained from $\log(1 - u) = -\sum_{k \geq 1} u^k/k$ is absolutely convergent when $|w| < 2$. Interchanging sums gives

$$\log \frac{\operatorname{sinc}(\pi w)}{1 - w^2} = -\sum_{k=1}^{\infty} \frac{w^{2k}}{k} \sum_{\ell=2}^{\infty} \ell^{-2k} = -\sum_{k=1}^{\infty} c_k w^{2k}.$$

Thus

$$\operatorname{sinc}(\pi w) = (1 - w^2) \exp\left(-\sum_{k=1}^{\infty} c_k w^{2k}\right), \quad |w| < 2. \quad (2.12)$$

Apply (2.12) with $w = z/2^n$ and multiply over $n \geq m$. The condition $|z| < 2^{m+1}$ places every factor inside the common disk of convergence. The algebraic factors give

$$\prod_{n=m}^{\infty} \left(1 - \frac{z^2}{4^n}\right) = (z^2/4^m; 1/4)_\infty.$$

For the exponent, absolute convergence allows us to interchange the n - and k -sums:

$$\begin{aligned} -\sum_{n=m}^{\infty} \sum_{k=1}^{\infty} c_k \frac{z^{2k}}{4^{nk}} &= -\sum_{k=1}^{\infty} c_k z^{2k} \frac{4^{-mk}}{1 - 4^{-k}} \\ &= -\sum_{k=1}^{\infty} \frac{c_k}{1 - 4^{-k}} \left(\frac{z}{2^m}\right)^{2k}. \end{aligned}$$

This proves (2.10); splitting the defining product at m proves (2.11). $\square$

Common pitfall

The disk $|z| < 2^{m+1}$ is the convergence disk of the chosen logarithmic expansion after extracting only the nearest zero pair. It is not the support of Φ , and it does not imply that Φ vanishes outside the disk. For a larger z , increase m , extract more zero layers, or analytically continue the formula.

3 Exact dyadic renormalization

All large-argument results in this article begin with one exact identity. It converts the infinite sinc product at a large point into a fixed compact “mantissa” factor and a finite lacunary sine product.

3.1 The master factorization

Let

$$T(t) = 2t \pmod{1}, \quad s(t) = |\sin(\pi t)|, \quad B_n(y) = \prod_{j=1}^n s(T^j y) = \prod_{j=1}^n |\sin(\pi 2^j y)|. \quad (3.1)$$

Theorem 3.1 (Exact dyadic factorization). *For every integer $n \geq 1$ and every $y \in [1/2, 1]$,*

$$|\Phi(2^n y)| = |\Phi(y)| \pi^{-n} y^{-n} 2^{-n(n+1)/2} B_n(y). \quad (3.2)$$

For a real κ , define the compact phase factor

$$W_\kappa(y) = |\Phi(y)| y^\kappa \exp\left(\frac{(\log y)^2}{2a}\right), \quad a = \log 2. \quad (3.3)$$

Then

$$\frac{|\Phi(2^n y)|}{E_\kappa(2^n y)} = W_\kappa(y) \exp(n(\kappa a - \log \pi - a/2)) B_n(y). \quad (3.4)$$

Proof. Iterating the first identity in (2.3) gives

$$\Phi(2^n y) = \Phi(y) \prod_{j=1}^n \text{sinc}(\pi 2^j y).$$

Taking absolute values separates each sinc factor into its numerator and denominator:

$$|\text{sinc}(\pi 2^j y)| = \frac{|\sin(\pi 2^j y)|}{\pi 2^j y}.$$

The product of the numerators is $B_n(y)$. The product of the denominators is

$$\prod_{j=1}^n (\pi 2^j y) = \pi^n y^n 2^{1+2+\dots+n} = \pi^n y^n 2^{n(n+1)/2}.$$

This proves (3.2).

To derive the normalized identity, put $L = \log(2^n y) = na + \log y$. Dividing by $E_\kappa(2^n y) = \exp(-L^2/(2a) - \kappa L)$ adds

$$\frac{L^2}{2a} + \kappa L = \frac{a}{2} n^2 + n \log y + \frac{(\log y)^2}{2a} + \kappa a n + \kappa \log y$$

to the logarithm. The term $n \log y$ cancels the factor y^{-n} , and $an^2/2$ cancels the quadratic part of $-an(n+1)/2$, leaving $-an/2$. The remaining terms are exactly (3.4). $\square$

3.2 Where the log-normal factor comes from

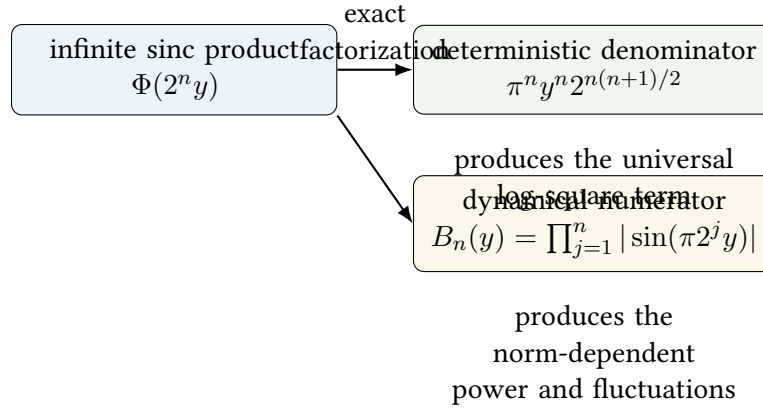
The origin of the leading decay can now be seen without any probability theory. The triangular denominator contributes

$$2^{-n(n+1)/2} = \exp\left(-\frac{a}{2}n^2 - \frac{a}{2}n\right).$$

Since $x = 2^n y$ with $y \in [1/2, 1]$, one has $n = (\log x - \log y)/a$. Substituting this into the quadratic term gives

$$-\frac{a}{2}n^2 = -\frac{(\log x)^2}{2a} + \frac{\log x \log y}{a} - \frac{(\log y)^2}{2a}. \quad (3.5)$$

The mixed term is canceled by y^{-n} , exactly as in the proof above. Thus $-(\log x)^2/(2 \log 2)$ is deterministic and unavoidable. The remaining finite product $B_n(y)$ decides only the coefficient of the linear term in n , hence the power of x , together with smaller fluctuations.



What the formula says

The difficult-looking entire function has been reduced to a familiar dynamical system: the doubling map on the circle. Different statistical questions about Φ become different statistical questions about the same cocycle $\log |\sin(\pi t)|$.

3.3 The Thue–Morse polynomial hidden in the numerator

Let

$$\tau_k = (-1)^{s_2(k)},$$

the signed Thue–Morse sequence. Define

$$P_n(t) = \prod_{j=0}^{n-1} \left(1 - e^{2\pi i 2^j t}\right). \quad (3.6)$$

Proposition 3.2 (Finite Thue–Morse expansion). *For every integer $n \geq 0$,*

$$P_n(t) = \sum_{k=0}^{2^n-1} \tau_k e^{2\pi i k t}, \quad |P_n(t)| = 2^n \prod_{j=0}^{n-1} s(T^j t). \quad (3.7)$$

Proof. The statement is immediate for $n = 0$. Assume the expansion for n . Then

$$\begin{aligned} P_{n+1}(t) &= P_n(t) (1 - e^{2\pi i 2^n t}) \\ &= \sum_{k=0}^{2^n-1} \tau_k e^{2\pi i k t} - \sum_{k=0}^{2^n-1} \tau_k e^{2\pi i (k+2^n) t}. \end{aligned}$$

The binary digit identities $\tau_k = (-1)^{s_2(k)}$ and $\tau_{k+2^n} = -\tau_k$ show that the two sums combine into the claimed expansion for $n + 1$. Finally, $|1 - e^{2\pi i u}| = 2|\sin \pi u|$, so taking moduli in (3.6) gives the second formula. $\square$

Thus the numerator in the Fourier decay problem is, up to a power of two, the modulus of a classical Thue–Morse trigonometric polynomial. Parseval, Gelfond's supremum estimate, Riesz products, and thermodynamic formalism are all available because of this exact bridge.

For completeness we also record the finite product-to-sum identity that motivated the original investigation.

Proposition 3.3 (Finite sinc product as a Thue–Morse sine sum). *For every integer $m \geq 0$ and every real $x \neq 0$,*

$$\prod_{h=0}^m \operatorname{sinc}\left(\frac{\pi x}{2^h}\right) = \frac{2^{m(m-1)/2}}{(\pi x)^{m+1}} \sum_{r=0}^{2^m-1} \tau_r \sin\left(\frac{\pi m}{2} + \frac{(2r+1)\pi x}{2^m}\right). \quad (3.8)$$

At $x = 0$, both sides have the continuous value one.

Proof. Put $\theta = \pi x/2^m$. Reversing the order of factors gives

$$\prod_{h=0}^m \sin\left(\frac{\pi x}{2^h}\right) = \prod_{j=0}^m \sin(2^j \theta).$$

Apply Theorem 3.2 with $m + 1$ factors and use

$$1 - e^{2iu} = -2ie^{iu} \sin u.$$

After solving for the product of sines, pair the term indexed by r with the term indexed by $2^{m+1} - 1 - r$. Binary complementation gives $\tau_{2^{m+1}-1-r} = (-1)^{m+1} \tau_r$, and taking the appropriate real or imaginary part yields

$$\prod_{j=0}^m \sin(2^j \theta) = 2^{-m} \sum_{r=0}^{2^m-1} \tau_r \sin\left(\frac{\pi m}{2} + (2r+1)\theta\right).$$

The product of the sinc denominators is

$$\prod_{h=0}^m \frac{\pi x}{2^h} = \frac{(\pi x)^{m+1}}{2^{m(m+1)/2}}.$$

Dividing the two formulas proves (3.8). Continuity at zero follows from the original product. $\square$

4 One product, many notions of size

The master identity reduces the choice of a decay exponent to one question: what is the exponential growth rate of B_n under the chosen functional?

For $q \in (0, \infty)$, define

$$\mathcal{M}_{q,n}(H) = \left(2 \int_{1/2}^1 |H(2^n y)|^q dy \right)^{1/q}. \quad (4.1)$$

This is identical to (1.3) after $x = 2^n y$. The endpoint cases are

$$\mathcal{M}_{0,n}(H) = \exp\left(2 \int_{1/2}^1 \log |H(2^n y)| dy \right), \quad (4.2)$$

$$\mathcal{M}_{\infty,n}(H) = \sup_{1/2 \leq y \leq 1} |H(2^n y)|. \quad (4.3)$$

Zeros cause no difficulty in (4.2) because logarithmic singularities are locally integrable.

Suppose, heuristically for the moment, that the chosen size of B_n behaves like Λ^n . In (3.4), the n -dependent exponential then has rate

$$\kappa a - \log \pi - a/2 + \log \Lambda.$$

To make the normalized size neither vanish nor blow up, we must set

$$\boxed{\kappa(\Lambda) = \frac{\log \pi + a/2 - \log \Lambda}{a}}. \quad (4.4)$$

Every exponent in the article is an instance of this one formula.

typical	mean	RMS	envelope	
$q = 0$	$q = 1$	$q = 2$	$q = \infty$	emphasis on large values
$\kappa \approx 3.1515$	$\kappa \approx 2.7481$	$\kappa \approx 2.6515$	$\kappa \approx 2.3590$	

The next four sections make the heuristic precise. The geometric-mean and RMS cases admit particularly elementary exact proofs. The general finite- q case is governed by a weighted transfer operator, and the supremum case is governed by an optimizing periodic orbit.

5 Geometric-mean and almost-everywhere decay

The $q = 0$ endpoint is elementary because Lebesgue measure is invariant under doubling and because the mean logarithm of the sine is known exactly.

Lemma 5.1 (The log-sine integral). *One has*

$$\int_0^1 \log |\sin(\pi t)| dt = -\log 2 = -a. \quad (5.1)$$

Proof. Let $I = \int_0^{\pi/2} \log(\sin u) du$. By $u \mapsto \pi/2 - u$, the same integral is $\int_0^{\pi/2} \log(\cos u) du$. Hence

$$\begin{aligned} 2I &= \int_0^{\pi/2} \log(\sin u \cos u) du \\ &= \int_0^{\pi/2} \log\left(\frac{1}{2} \sin 2u\right) du = -\frac{\pi}{2} \log 2 + \frac{1}{2} \int_0^{\pi} \log(\sin v) dv. \end{aligned}$$

Symmetry gives $\int_0^{\pi} \log(\sin v) dv = 2I$, so $I = -(\pi/2) \log 2$. Rescaling and using symmetry once more gives (5.1). $\square$

Theorem 5.2 (Exact shell geometric mean). *Set*

$$\boxed{\kappa_0 = \frac{3}{2} + \frac{\log \pi}{a}} = 3.151496129472318798 \dots \quad (5.2)$$

Then, for every $n \geq 1$,

$$\mathcal{M}_{0,n}\left(\frac{\Phi}{E_{\kappa_0}}\right) = \exp\left(2 \int_{1/2}^1 \log W_{\kappa_0}(y) dy\right). \quad (5.3)$$

In particular, the normalized shell geometric mean is exactly independent of n .

Proof. Take logarithms in (3.4), multiply by two, and integrate over $y \in [1/2, 1]$. Put $z = 2y - 1$. For $j \geq 1$, $T^j y = T^{j-1} z$, and $2 dy = dz$. Since Lebesgue measure is invariant under T , Theorem 5.1 gives

$$2 \int_{1/2}^1 \log B_n(y) dy = \sum_{j=1}^n \int_0^1 \log s(T^{j-1} z) dz = -na.$$

For $\kappa = \kappa_0$,

$$\kappa_0 a - \log \pi - a/2 = a.$$

Thus the deterministic contribution $+na$ in (3.4) cancels the integral $-na$ of the numerator exactly. Only the compact phase factor remains, proving (5.3). $\square$

The same exponent governs almost every fixed dyadic ray.

Theorem 5.3 (Typical fixed-mantissa law). *For Lebesgue-almost every $y \in [1/2, 1]$,*

$$\log |\Phi(2^n y)| = -\frac{(\log(2^n y))^2}{2a} - \kappa_0 \log(2^n y) + o(n). \quad (5.4)$$

Equivalently,

$$|\Phi(2^n y)| = E_{\kappa_0}(2^n y) e^{o(n)}. \quad (5.5)$$

Proof. The function $\log s$ belongs to $L^1(0, 1)$, and the doubling map is ergodic with respect to Lebesgue measure. Birkhoff's ergodic theorem therefore gives, for almost every y ,

$$\frac{1}{n} \log B_n(y) = \frac{1}{n} \sum_{j=1}^n \log s(T^j y) \longrightarrow \int_0^1 \log s(t) dt = -a.$$

Thus $\log B_n(y) = -an + o(n)$. Substitute this into (3.4); the choice $\kappa = \kappa_0$ contributes $+an$, so the linear terms cancel and the logarithm of the normalized quotient is $o(n)$. $\square$

Remark 5.4 (Exceptional phases). The qualifier “almost every” is essential. If y is dyadic, then some iterate $T^j y$ is zero and $B_n(y) = 0$ thereafter. Periodic non-dyadic points have Birkhoff averages determined by their finite orbit, which need not equal $-\log 2$. The sharp uniform envelope will come from the period-two orbit $1/3 \leftrightarrow 2/3$, not from a typical orbit.

6 Typical fluctuations: exact variance, CLT, and LIL

The error $o(n)$ in Theorem 5.3 is not a bounded remainder. It has Gaussian fluctuations of order $\sqrt{n}$, and almost-sure extreme fluctuations of order $\sqrt{n \log \log n}$. In this problem both the variance and the martingale decomposition are explicit.

Put

$$\psi(t) = \log |2 \sin(\pi t)|, \quad S_n(t) = \sum_{j=0}^{n-1} \psi(T^j t). \quad (6.1)$$

Since $\log s = \psi - a$, the choice κ_0 turns the master identity into

$$\log \frac{|\Phi(2^n y)|}{E_{\kappa_0}(2^n y)} = \log W_{\kappa_0}(y) + \sum_{j=1}^n \psi(T^j y). \quad (6.2)$$

Thus all nontrivial fluctuations are carried by S_n .

6.1 Fourier expansion and exact covariance

Lemma 6.1 (Fourier series of the log-sine cocycle). *In $L^2(0, 1)$,*

$$\psi(t) = - \sum_{m=1}^{\infty} \frac{\cos(2\pi mt)}{m}. \quad (6.3)$$

In particular, $\int_0^1 \psi = 0$.

Proof. For $0 < r < 1$, the power series for the logarithm gives

$$-\Re \log(1 - re^{2\pi it}) = \sum_{m=1}^{\infty} \frac{r^m \cos(2\pi mt)}{m}.$$

As $r \uparrow 1$, the left side tends almost everywhere to $-\log|1 - e^{2\pi it}| = -\log|2 \sin \pi t| = -\psi(t)$. The Fourier coefficients $1/m$ are square summable, so the right side converges in L^2 to the series in (6.3). The constant Fourier coefficient is zero, agreeing with [Theorem 5.1](#) after the added $\log 2$. $\square$

Theorem 6.2 (Exact covariance and finite-level variance). *For every integer $r \geq 0$,*

$$C_r := \int_0^1 \psi(t) \psi(T^r t) dt = \frac{\pi^2}{12 \cdot 2^r}. \quad (6.4)$$

Consequently, for every $n \geq 1$,

$$\boxed{\int_0^1 S_n(t)^2 dt = \frac{\pi^2}{4} n - \frac{\pi^2}{3} (1 - 2^{-n})}. \quad (6.5)$$

The asymptotic variance per dyadic level is $\sigma^2 = \pi^2/4$.

Proof. For $r = 0$, Parseval's identity and $\int_0^1 \cos^2(2\pi mt) dt = 1/2$ give

$$C_0 = \frac{1}{2} \sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{\pi^2}{12}.$$

For $r \geq 1$, substitute the Fourier series for both factors. The frequencies in $\psi(T^r t)$ are $2^r \ell$. Orthogonality leaves only pairs with $m = 2^r \ell$, hence

$$C_r = \frac{1}{2} \sum_{\ell=1}^{\infty} \frac{1}{(2^r \ell) \ell} = \frac{\pi^2}{12 \cdot 2^r}.$$

Stationarity of Lebesgue measure under T now yields

$$\begin{aligned} \int_0^1 S_n^2 &= nC_0 + 2 \sum_{r=1}^{n-1} (n-r)C_r \\ &= \frac{\pi^2}{12} \left[n + 2 \sum_{r=1}^{n-1} (n-r)2^{-r} \right]. \end{aligned}$$

The elementary finite sum is

$$\sum_{r=1}^{n-1} (n-r)2^{-r} = n - 2 + 2^{1-n}.$$

Substitution simplifies to (6.5). Dividing by n and letting $n \rightarrow \infty$ gives $\sigma^2 = \pi^2/4$. $\square$

Common pitfall

The factor $1/2$ in $\int_0^1 \cos^2(2\pi mt) dt = 1/2$ is essential. Omitting it doubles the variance and therefore enlarges the LIL constant by $\sqrt{2}$.

6.2 An exact martingale-coboundary decomposition

Let $\mathcal{P}$ be the unweighted Perron operator of the doubling map:

$$(\mathcal{P}f)(x) = \frac{1}{2} (f(x/2) + f((x+1)/2)). \quad (6.6)$$

Lemma 6.3 (Gordin decomposition in closed form). *One has*

$$\mathcal{P}\psi = \frac{1}{2}\psi, \quad d := 2\psi - \psi \circ T = \log |\tan(\pi t)|, \quad \mathcal{P}d = 0, \quad (6.7)$$

and the exact telescoping identity

$$S_n = \sum_{j=0}^{n-1} d \circ T^j - \psi + \psi \circ T^n. \quad (6.8)$$

Moreover,

$$\int_0^1 d(t)^2 dt = \frac{\pi^2}{4}. \quad (6.9)$$

Proof. The action of $\mathcal{P}$ on Fourier modes is

$$\mathcal{P}e^{2\pi i m t} = \begin{cases} e^{2\pi i (m/2)t}, & m \text{ even}, \\ 0, & m \text{ odd}. \end{cases}$$

In the series (6.3), only the even frequencies survive; replacing $m = 2\ell$ gives $\mathcal{P}\psi = \psi/2$. Also $\mathcal{P}(f \circ T) = f$, so $\mathcal{P}d = 2\mathcal{P}\psi - \mathcal{P}(\psi \circ T) = \psi - \psi = 0$. The double-angle formula gives

$$2\psi(t) - \psi(2t) = \log \frac{|2 \sin \pi t|^2}{|2 \sin 2\pi t|} = \log |\tan \pi t|,$$

which proves the middle identity in (6.7). Summing $d \circ T^j = 2\psi \circ T^j - \psi \circ T^{j+1}$ from $j = 0$ to $n-1$ and collecting terms gives (6.8).

For the variance, symmetry and the substitution $u = \pi t$ give

$$\int_0^1 \log^2 |\tan \pi t| dt = \frac{2}{\pi} \int_0^{\pi/2} \log^2(\tan u) du.$$

For $|s| < 1$, Euler's beta integral yields

$$F(s) = \int_0^{\pi/2} \tan^s u du = \frac{1}{2} \Gamma\left(\frac{1+s}{2}\right) \Gamma\left(\frac{1-s}{2}\right) = \frac{\pi}{2 \cos(\pi s/2)}.$$

Differentiating twice at $s = 0$ is justified by dominated convergence on every compact subinterval of $(-1, 1)$. We obtain

$$\int_0^{\pi/2} \log^2(\tan u) du = F''(0) = \frac{\pi^3}{8}.$$

Multiplication by $2/\pi$ proves (6.9). □

The condition $\mathcal{P}d = 0$ means that $d \circ T^j$ is a stationary reverse martingale-difference sequence for the decreasing filtration generated by future binary digits. The two remaining terms in (6.8) are a coboundary.

6.3 Limit laws

We use two standard probability theorems. Brown's martingale-array central limit theorem [4] gives the Gaussian limit after a finite reverse martingale row is read in the opposite order. Cuny's compact law of the iterated logarithm [5] applies directly to stationary ergodic reverse martingale differences. We verify all product-specific hypotheses below.

Theorem 6.4 (Central limit theorem and corrected LIL). *With respect to Lebesgue measure on $(0, 1)$,*

$$\frac{S_n}{\sqrt{n}} \Longrightarrow N\left(0, \frac{\pi^2}{4}\right). \quad (6.10)$$

For almost every t ,

$$\limsup_{n \rightarrow \infty} \frac{S_n(t)}{\sqrt{2n \log \log n}} = \frac{\pi}{2}, \quad \liminf_{n \rightarrow \infty} \frac{S_n(t)}{\sqrt{2n \log \log n}} = -\frac{\pi}{2}. \quad (6.11)$$

Equivalently, with denominator $\sqrt{n \log \log n}$, the constants are $\pm\pi/\sqrt{2}$.

Proof. Let $\mathcal{F}_j = T^{-j}\mathcal{B}$, a decreasing filtration. The identity $\mathcal{P}d = 0$ is equivalent to

$$\mathbb{E}[d \circ T^j \mid \mathcal{F}_{j+1}] = 0.$$

Thus $(d \circ T^j)_{j \geq 0}$ is a stationary reverse martingale-difference sequence. The logarithmic singularities of $d = \log |\tan \pi t|$ are mild: there is a constant C such that

$$\text{Leb}\{|d| > u\} \leq Ce^{-u}, \quad u \geq 1.$$

Indeed, $|\tan \pi t| < e^{-u}$ confines t to intervals of total length $O(e^{-u})$ around the zeros of the tangent, and $|\tan \pi t| > e^u$ does the same around its poles. Hence $d \in L^p$ for every finite p , in particular $d \in L^2$.

For the central limit theorem, fix n and read the finite reverse-martingale row in the opposite order:

$$X_{n,k} = n^{-1/2} d \circ T^{n-k}, \quad \mathcal{G}_{n,k} = \mathcal{F}_{n-k}, \quad 1 \leq k \leq n.$$

Because $(\mathcal{F}_j)$ is decreasing, $(\mathcal{G}_{n,k})_{k=0}^n$ is increasing, and

$$\mathbb{E}[X_{n,k} \mid \mathcal{G}_{n,k-1}] = 0.$$

The sum of the row is exactly $n^{-1/2} \sum_{j=0}^{n-1} d \circ T^j$; only the order of its summands has changed. Its predictable quadratic variation is

$$\sum_{k=1}^n \mathbb{E}[X_{n,k}^2 \mid \mathcal{G}_{n,k-1}] = \frac{1}{n} \sum_{j=0}^{n-1} (\mathcal{P}d^2) \circ T^{j+1}.$$

By the ergodic theorem this converges almost surely to $\int d^2 = \pi^2/4$. Moreover, for every $\varepsilon > 0$,

$$\sum_{k=1}^n \mathbb{E}\left[X_{n,k}^2 \mathbf{1}_{\{|X_{n,k}| > \varepsilon\}} \mid \mathcal{G}_{n,k-1}\right] \longrightarrow 0$$

in probability. Indeed, the expectation of this nonnegative sum is exactly

$$\int_0^1 d^2 \mathbf{1}_{\{|d| > \varepsilon \sqrt{n}\}} dt,$$

which tends to zero by dominated convergence; Markov's inequality then gives the asserted conditional Lindeberg condition. Brown's martingale central limit theorem therefore gives

$$n^{-1/2} \sum_{j=0}^{n-1} d \circ T^j \Longrightarrow N(0, \pi^2/4).$$

For the iterated-logarithm law, no time-reversal argument is needed. The scalar specialization of Cuny's compact LIL applies directly to the stationary ergodic reverse martingale-difference sequence $(d \circ T^j)$: it is square-integrable and has variance $\|d\|_2^2 = \pi^2/4$. In one dimension, the compact LIL says that the set of limit points of

$$\frac{\sum_{j=0}^{n-1} d \circ T^j}{\sqrt{2n \log \log n}}$$

is the interval $[-\|d\|_2, \|d\|_2]$ almost surely. Hence the martingale sum has $\limsup \pi/2$ and $\liminf -\pi/2$.

It remains to remove the coboundary in (6.8). The positive part of ψ is bounded by $\log 2$, while $\text{Leb}\{\psi < -u\} = O(e^{-u})$. Therefore, for every $\varepsilon > 0$,

$$\sum_{n=3}^{\infty} \text{Leb} \left\{ |\psi \circ T^n| > \varepsilon \sqrt{n \log \log n} \right\} < \infty.$$

Borel–Cantelli gives $\psi \circ T^n = o(\sqrt{n \log \log n})$ almost surely. The same tail bound, with threshold $\varepsilon \sqrt{n}$, makes the coboundary $o(\sqrt{n})$ in probability. Thus it does not change either the CLT or the LIL. $\square$

Combining (6.2) with Theorem 6.4 describes the fluctuations of the Fourier transform on almost every fixed ray. In terms of x , the extremal multiplicative fluctuation has the form

$$\exp \left[\pm \left(\frac{\pi}{\sqrt{2}} + o(1) \right) \sqrt{\frac{\log x}{\log 2} \log \log \log x} \right]. \quad (6.12)$$

It is subpolynomial in x , but it is larger than every fixed power of $\log x$.

Remark 6.5 (Normalization check against the literature). Aistleitner–Hofer–Larcher [1] study the same lacunary product. Their printed LIL constant with the normalization $\sqrt{n \log \log n}$ is larger by $\sqrt{2}$ than (6.11). The exact covariance computation above locates the normalization issue: the cosine basis has squared $L^2(0, 1)$ -norm $1/2$, not one. None of the exponential decay exponents depends on this correction.

7 The transfer operator and the finite- q spectrum

The geometric mean used only the integral of $\log s$. For an arithmetic mean or a general L^q mean, one must understand the weighted products themselves. The correct bookkeeping device is a Perron–Frobenius operator.

7.1 Derivation of the weighted operator

After the substitution $x = 2y - 1$, one has $Ty = x$ and therefore

$$B_n(y) = \prod_{j=0}^{n-1} s(T^j x), \quad y = \frac{x+1}{2}. \quad (7.1)$$

For $q > 0$, define

$$(\mathcal{L}_q f)(x) = \frac{1}{2} \left[\sin^q \left(\frac{\pi x}{2} \right) f(x/2) + \cos^q \left(\frac{\pi x}{2} \right) f((x+1)/2) \right]. \quad (7.2)$$

The sine and cosine are nonnegative on the displayed interval, so no absolute-value symbols are needed in (7.2).

Lemma 7.1 (Transfer duality). *For every integrable f and every integer $n \geq 0$,*

$$\int_0^1 f(x) \prod_{j=0}^{n-1} s(T^j x)^q dx = \int_0^1 (\mathcal{L}_q^n f)(x) dx. \quad (7.3)$$

Proof. For $n = 1$, split the integral at $1/2$. On the first half put $x = u/2$; on the second put $x = (u+1)/2$. Since

$$s(u/2) = \sin(\pi u/2), \quad s((u+1)/2) = \cos(\pi u/2),$$

we obtain

$$\int_0^1 f(x) s(x)^q dx = \int_0^1 (\mathcal{L}_q f)(u) du.$$

Apply this identity successively to the last, then the next-to-last, factor in the product. This proves (7.3) by induction. $\square$

7.2 The precise spectral input

The next proposition is a standard Ruelle–Perron–Frobenius theorem specialized to the operator (7.2). We state it explicitly because every later use depends only on its rank-one asymptotic. The product-specific work consists in checking the hypotheses; the general quasi-compactness theorem is imported from transfer-operator theory [2, 6].

Proposition 7.2 (Ruelle–Perron–Frobenius decomposition). *Fix $q > 0$, and choose $0 < \alpha \leq \min(1, q)$. On the Banach space $C^\alpha[0, 1]$, the operator $\mathcal{L}_q$ has a simple leading eigenvalue $\rho_q > 0$, a strictly positive eigenfunction h_q , and a positive eigenmeasure ν_q , normalized by*

$$\mathcal{L}_q h_q = \rho_q h_q, \quad \mathcal{L}_q^* \nu_q = \rho_q \nu_q, \quad \nu_q(h_q) = 1. \quad (7.4)$$

There are constants $C > 0$ and $0 < \vartheta < 1$ such that

$$\|\rho_q^{-n} \mathcal{L}_q^n f - \nu_q(f) h_q\|_{C^\alpha} \leq C \vartheta^n \|f\|_{C^\alpha} \quad (n \geq 0). \quad (7.5)$$

Why the standard theorem applies. Write the inverse branches of doubling as $\iota_0(x) = x/2$ and $\iota_1(x) = (x+1)/2$, with weights

$$w_0(x) = \frac{1}{2} \sin^q(\pi x/2), \quad w_1(x) = \frac{1}{2} \cos^q(\pi x/2).$$

For $0 < \alpha \leq \min(1, q)$, both weights belong to $C^\alpha[0, 1]$, and both inverse branches contract distances by $1/2$. Thus $\mathcal{L}_q$ is a finite-branch Hölder transfer operator for a uniformly expanding map, exactly the class covered by the cited Ruelle–Perron–Frobenius theorem.

It remains to check that the two endpoint zeros of the weights do not create an invariant component. Let $f \geq 0$ be continuous and not identically zero. It is positive on some open interval U . Choose n so large that the mesh 2^{-n} is much smaller than the length of U . For every $x \in [0, 1]$, the n -fold preimages $(x+k)/2^n$, $0 \leq k < 2^n$, then include a point of U ; because U contains many such mesh points, one can avoid the finitely many dyadic branch endpoints at which a cylinder weight vanishes. The corresponding summand in $\mathcal{L}_q^n f(x)$ is positive. Thus one and the same iterate sends every nonzero nonnegative continuous f into the interior of the positive cone, proving primitivity.

The nonnegative-weight form of the standard RPF theorem now gives quasi-compactness, a positive eigenfunction and eigenmeasure at the spectral radius, simplicity of that eigenvalue, absence of other peripheral spectrum, and the rank-one estimate (7.5); see [2, 6]. This paragraph verifies the hypotheses of that theorem. It is not intended as an independent proof of the general spectral-gap theorem, which is the functional-analytic input explicitly imported here. $\square$

Proof status

The remainder of the finite- q derivation is elementary once (7.5) is granted. Thus the functional-analytic input is isolated in one clearly marked proposition rather than being silently invoked throughout the article.

7.3 Sharp finite- q shell asymptotics

Define

$$\Lambda_q = \rho_q^{1/q}, \quad \kappa_q = \frac{\log \pi + a/2 - \log \Lambda_q}{a}. \quad (7.6)$$

Theorem 7.3 (Finite- q decay spectrum). *For every fixed $q \in (0, \infty)$, there is $A_q \in (0, \infty)$ such that*

$$\mathcal{M}_{q,n} \left(\frac{\Phi}{E_{\kappa_q}} \right) \rightarrow A_q. \quad (7.7)$$

If

$$F_q(x) = W_{\kappa_q} \left(\frac{x+1}{2} \right)^q, \quad (7.8)$$

then, with the normalization (7.4),

$$A_q^q = \left(\int_0^1 h_q(x) dx \right) \nu_q(F_q). \quad (7.9)$$

Moreover, $q \mapsto \Lambda_q$ is nondecreasing and $q \mapsto \kappa_q$ is nonincreasing.

Proof. Raise (3.4) to the q th power and use (4.1). With $x = 2y - 1$ and (7.1),

$$\begin{aligned} \mathcal{M}_{q,n} \left(\frac{\Phi}{E_{\kappa}} \right)^q &= \exp(qn(\kappa a - \log \pi - a/2)) \\ &\times \int_0^1 W_{\kappa} \left(\frac{x+1}{2} \right)^q \prod_{j=0}^{n-1} s(T^j x)^q dx. \end{aligned} \quad (7.10)$$

The test function belongs to C^α for $0 < \alpha \leq \min(1, q)$: its only possible endpoint issue is the zero of Φ at one, and that zero has finite order. By Theorem 7.1 and (7.5), the integral in (7.10) equals

$$\rho_q^n \left(\int_0^1 h_q \right) \nu_q(W_{\kappa}((\cdot + 1)/2)^q) + O((\rho_q \vartheta)^n).$$

For $\kappa = \kappa_q$, definition (7.6) is precisely the cancellation condition

$$q(\kappa_q a - \log \pi - a/2) + \log \rho_q = 0.$$

The leading exponential disappears, and (7.7)–(7.9) follow. Positivity of A_q follows from positivity of h_q , ν_q , and the nonzero test function.

For each fixed n , the L^q norm of $\prod_{j=0}^{n-1} s \circ T^j$ on the probability space $[0, 1]$ is nondecreasing in q . Taking n th roots and then using the spectral limit shows that Λ_q is nondecreasing. Formula (7.6) reverses this ordering for κ_q . $\square$

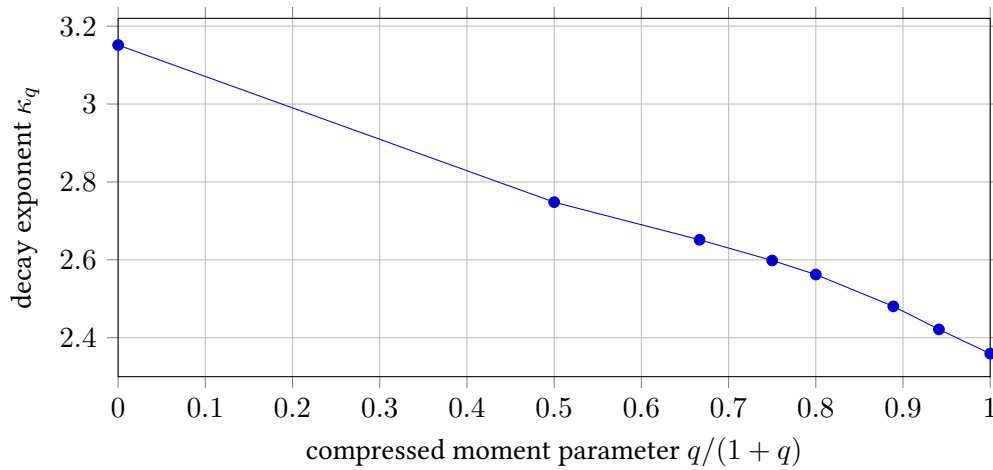
Corollary 7.4 (No logarithmic correction at the sharp power). *Fix $q \in (0, \infty)$. Replacing $E_{\kappa_q}(x)$ by $E_{\kappa_q}(x)(\log x)^p$ multiplies the normalized shell mean by $(na)^{-p}(1 + o(1))$. Hence the only logarithmic exponent compatible with a finite nonzero limit is $p = 0$.*

Proof. On $[2^{n-1}, 2^n]$, $\log x = na + O(1)$ uniformly. Pulling the slowly varying factor out of the normalized shell norm gives the stated multiplier, and its limit is zero for $p > 0$, infinite for $p < 0$, and one for $p = 0$. $\square$

7.4 Numerical orientation

The endpoint values $q = 0, 2, \infty$ are exact. For other q , the leading eigenvalue can be computed by Chebyshev collocation or interval transfer bounds. The following table is included only to display the shape of the spectrum; rows not marked exact are numerical.

q	ρ_q	$\Lambda_q = \rho_q^{1/q}$	κ_q	
0	—	0.5000000000	3.1514961295	exact
1	0.6613226021	0.6613226021	2.7480700149	
2	0.5000000000	0.7071067812	2.6514961295	exact
3	0.3948966324	0.7336593838	2.5983138062	
4	0.3201941016	0.7522346457	2.5622414703	
8	0.1612444589	0.7960412993	2.4805809434	
16	0.0500719357	0.8293247927	2.4214870020	
∞	—	0.8660254038	2.3590148791	exact



8 The exact root-mean-square exponent

The case $q = 2$ collapses to finite Fourier algebra; no numerical spectral computation is needed.

Theorem 8.1 (Exact quadratic lacunary identity). *For every integer $n \geq 0$,*

$$\int_0^1 \prod_{j=0}^{n-1} \sin^2(\pi 2^j t) dt = 2^{-n}. \quad (8.1)$$

Equivalently, $\rho_2 = 1/2$ and $\Lambda_2 = 2^{-1/2}$.

Proof. By Theorem 3.2,

$$P_n(t) = \sum_{k=0}^{2^n-1} \tau_k e^{2\pi i k t}, \quad |P_n(t)| = 2^n \prod_{j=0}^{n-1} |\sin(\pi 2^j t)|.$$

Parseval's identity for this finite Fourier polynomial gives

$$\int_0^1 |P_n(t)|^2 dt = \sum_{k=0}^{2^n-1} |\tau_k|^2 = 2^n.$$

Dividing by 4^n proves (8.1).

Alternatively, the transfer operator satisfies

$$\mathcal{L}_2 \mathbf{1} = \frac{1}{2} \left(\sin^2 \frac{\pi x}{2} + \cos^2 \frac{\pi x}{2} \right) = \frac{1}{2} \mathbf{1}.$$

Its positive leading eigenvalue is therefore $1/2$, in agreement with the finite identity. $\square$

Corollary 8.2 (RMS decay). *The sharp root-mean-square exponent is*

$$\kappa_2 = \frac{\log(2\pi)}{\log 2} = 2.651496129472318798 \dots \quad (8.2)$$

There is $A_2 \in (0, \infty)$ such that

$$\mathcal{M}_{2,n} \left(\frac{\Phi}{E_{\kappa_2}} \right) \rightarrow A_2. \quad (8.3)$$

Proof. In (4.4), insert $\Lambda = 2^{-1/2}$:

$$\kappa_2 = \frac{\log \pi + a/2 + a/2}{a} = \frac{\log(2\pi)}{a}.$$

The convergence statement is the $q = 2$ case of Theorem 7.3. $\square$

8.1 A discrete Parseval companion

Proposition 8.3 (Dyadic-grid identity). *For every $n \geq 1$, if $Q_n(t) = \prod_{j=0}^{n-1} |\sin(\pi 2^j t)|$, then*

$$\sum_{m=0}^{2^n-1} Q_n(m/2^n)^2 = 1. \quad (8.4)$$

Proof. Let $N = 2^n$. Discrete Parseval for the values of the degree- $(N-1)$ polynomial P_n on the N th roots of unity gives

$$\sum_{m=0}^{N-1} |P_n(m/N)|^2 = N \sum_{k=0}^{N-1} |\tau_k|^2 = N^2.$$

Since $|P_n(t)| = N Q_n(t)$, division by N^2 proves the claim. $\square$

8.2 Total Fourier energy and half-integer sampling

Proposition 8.4 (Energy sampling identity). *One has*

$$\int_{\mathbb{R}} |\Phi(x)|^2 dx = \int_{\mathbb{R}} \text{up}(t)^2 dt = \frac{1}{2} + \sum_{m=0}^{\infty} \Phi\left(m + \frac{1}{2}\right)^2. \quad (8.5)$$

Numerically, the common value is $0.8088735359679705629 \dots$; this decimal is numerical evidence, not an interval certificate.

Proof. The first equality is Plancherel's theorem under the convention (1.1). Since up is supported in $[-1, 1]$, expand it as a Fourier series of period two on that interval. Its k th coefficient is

$$c_k = \frac{1}{2} \int_{-1}^1 \text{up}(t) e^{-\pi i k t} dt = \frac{1}{2} \Phi(k/2).$$

Parseval on an interval of length two gives

$$\int_{-1}^1 \text{up}(t)^2 dt = 2 \sum_{k \in \mathbb{Z}} |c_k|^2 = \frac{1}{2} \sum_{k \in \mathbb{Z}} \Phi(k/2)^2.$$

The $k = 0$ term contributes $1/2$. Every nonzero even k samples Φ at a nonzero integer and therefore vanishes by Theorem 2.3. The positive and negative odd terms pair by evenness. Writing $k = \pm(2m+1)$ yields (8.5). $\square$

9 Arithmetic-mean decay and ordinary Cesàro averaging

For $q = 1$, the transfer operator is

$$(\mathcal{L}_1 f)(x) = \frac{1}{2} \left[\sin\left(\frac{\pi x}{2}\right) f(x/2) + \cos\left(\frac{\pi x}{2}\right) f((x+1)/2) \right]. \quad (9.1)$$

Its Perron root has no known elementary closed form, but it is a canonical spectral constant and admits short exact enclosures.

9.1 The Perron root and an exact rational certificate

High-precision spectral computation gives

$$\rho_1 = 0.661322602060565615\dots, \quad \boxed{\kappa_1 = \frac{\log \pi + a/2 - \log \rho_1}{a}} = 2.748070014871332674\dots \quad (9.2)$$

The decimal is numerical. The following enclosure, by contrast, is exact.

Proposition 9.1 (Rational Collatz–Wielandt enclosure). *One has*

$$0.66126807 < \rho_1 < 0.66134891, \quad (9.3)$$

and hence

$$2.74801262 < \kappa_1 < 2.74818899. \quad (9.4)$$

Proof. For any continuous $f > 0$, the eigenmeasure equation in (7.4) implies the Collatz–Wielandt bounds

$$\min_{x \in [0,1]} \frac{\mathcal{L}_1 f(x)}{f(x)} \leq \rho_1 \leq \max_{x \in [0,1]} \frac{\mathcal{L}_1 f(x)}{f(x)}. \quad (9.5)$$

Indeed, if $m f \leq \mathcal{L}_1 f \leq M f$, integration against ν_1 gives $m \nu_1(f) \leq \rho_1 \nu_1(f) \leq M \nu_1(f)$, and $\nu_1(f) > 0$.

Choose

$$p(u) = 1 + \frac{376189}{10^6} u - \frac{69093}{10^6} u^2 + \frac{15483}{10^6} u^3, \quad f(x) = p(\sin \pi x). \quad (9.6)$$

The polynomial is positive on $[0, 1]$, since $p(u) \geq 1 - 69093/10^6 > 0$. Put

$$t = \tan \frac{\pi x}{4}, \quad u = \sin \frac{\pi x}{2} = \frac{2t}{1+t^2}, \quad v = \cos \frac{\pi x}{2} = \frac{1-t^2}{1+t^2}.$$

As x runs from zero to one, t runs from zero to one, and $\sin \pi x = 2uv$. Therefore

$$\frac{\mathcal{L}_1 f(x)}{f(x)} = R(t) := \frac{u p(u) + v p(v)}{2p(2uv)}, \quad 0 \leq t \leq 1. \quad (9.7)$$

After clearing the positive denominator, the inequalities

$$\frac{66126807}{10^8} < R(t) < \frac{66134891}{10^8}$$

become positivity of two explicit integer polynomials of degree twelve. Their Sturm chains have the same number of sign variations at 0 and 1, namely six, and both polynomials are positive at zero. Hence neither has a root on $[0, 1]$, and both remain positive there. The complete exact integer computation is reproduced in [Section B](#); no floating-point step enters this sign proof. Applying (9.5) gives (9.3). Formula (9.4) follows from the strictly decreasing map $r \mapsto (\log \pi + a/2 - \log r)/a$. $\square$

Remark 9.2. Aistleitner–Hofer–Larcher [1] give the sharper rigorous interval $0.66130 < \rho_1 < 0.66135$ by a transfer recursion. The slightly wider bracket above is retained because it can be checked from one small polynomial and exact Sturm arithmetic.

9.2 The natural arithmetic-mean gauge

By Theorem 7.3,

$$2 \int_{1/2}^1 \frac{|\Phi(2^n y)|}{E_{\kappa_1}(2^n y)} dy \longrightarrow A_1, \quad (9.8)$$

where the exact spectral expression is

$$A_1 = \left(\int_0^1 h_1(x) dx \right) \nu_1(W_{\kappa_1}((\cdot + 1)/2)). \quad (9.9)$$

Numerically,

$$A_1 = 0.091266124131588 \dots \quad (9.10)$$

Define

$$g_1(x) = A_1 E_{\kappa_1}(x). \quad (9.11)$$

Then the mean of $|\Phi|/g_1$ on every complete dyadic shell tends to one.

Proposition 9.3 (Regularity of the natural gauge). *The gauge g_1 is positive and real analytic on $(0, \infty)$, and it is strictly decreasing for $x \geq 1$. For every fixed integer $m \geq 0$, there is X_m such that*

$$(-1)^m g_1^{(m)}(x) > 0, \quad x \geq X_m. \quad (9.12)$$

Proof. Put $u = \log x$ and

$$G(u) = g_1(e^u) = A_1 \exp\left(-\frac{u^2}{2a} - \kappa_1 u\right).$$

Analyticity and positivity are immediate. Directly,

$$\frac{g_1'(x)}{g_1(x)} = -\frac{u/a + \kappa_1}{x} < 0$$

for $x \geq 1$. For higher derivatives use

$$x^m \frac{d^m}{dx^m} = D_u(D_u - 1) \cdots (D_u - m + 1), \quad D_u = \frac{d}{du}.$$

Let $P(u) = u/a + \kappa_1$. Since $D_u G = -P(u)G$, induction on m gives

$$(-1)^m \frac{x^m g_1^{(m)}(x)}{g_1(x)} = P(u)^m + O_m(P(u)^{m-1}). \quad (9.13)$$

The leading term is positive and dominates as $u \rightarrow \infty$, proving (9.12). $\square$

Corollary 9.4 (Dyadic Cesàro convergence and unrestricted boundedness). *For every fixed $t_0 > 0$,*

$$\frac{1}{2^n - t_0} \int_{t_0}^{2^n} \frac{|\Phi(x)|}{g_1(x)} dx \longrightarrow 1. \quad (9.14)$$

Moreover,

$$0 < \liminf_{t \rightarrow \infty} \frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g_1(x)} dx \leq \limsup_{t \rightarrow \infty} \frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g_1(x)} dx < \infty. \quad (9.15)$$

Proof. Let

$$a_n = \frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} \frac{|\Phi(x)|}{g_1(x)} dx.$$

Equation (9.8) says $a_n \rightarrow 1$. Therefore

$$\int_1^{2^N} \frac{|\Phi(x)|}{g_1(x)} dx = \sum_{n=1}^N 2^{n-1} a_n.$$

A weighted Cesàro argument gives $2^{-N} \sum_{n=1}^N 2^{n-1} a_n \rightarrow 1$: split the sum at a fixed index after which $|a_n - 1| < \varepsilon$, and note that the finite initial part is $o(2^N)$. This proves (9.14); changing the fixed lower endpoint contributes only $O(1)/2^N$.

If $2^{N-1} \leq t \leq 2^N$, the completed shells up to 2^{N-1} give a lower bound asymptotic to 2^{N-1} , while including the entire current shell gives an upper bound asymptotic to 2^N . Since t lies between those same quantities, division by t gives positive finite uniform lower and upper bounds. Again, a fixed t_0 is negligible. $\square$

9.3 The endpoint phase left by an incomplete shell

The previous corollary does *not* imply an unrestricted limit. A final incomplete shell contributes a fixed fraction of the total additive average, and its normalized mass converges to a singular distribution.

Let

$$f_1(x) = W_{\kappa_1} \left(\frac{x+1}{2} \right), \quad D(z) = \frac{\nu_1(f_1 \mathbf{1}_{[0,z]})}{\nu_1(f_1)}, \quad 0 \leq z \leq 1. \quad (9.16)$$

Lemma 9.5 (The limiting shell measure is continuous and singular). *The probability measure*

$$\mu(A) = \frac{\nu_1(f_1 \mathbf{1}_A)}{\nu_1(f_1)} \quad (9.17)$$

is atomless, fully supported, and singular with respect to Lebesgue measure. Consequently D is a continuous, nondecreasing singular distribution function with $D(0) = 0$ and $D(1) = 1$.

Proof. Primitivity in the proof of Theorem 7.2 implies that ν_1 gives positive mass to every nonempty open interval. Since f_1 is continuous and positive on $[0, 1]$, the same is true for μ .

We next rule out atoms. The transfer weights at the endpoints show directly from $\mathcal{L}_1^* \nu_1 = \rho_1 \nu_1$ that $\nu_1(\{0\}) = \nu_1(\{1\}) = 0$. The possible atom at $1/2$ is fed only by these endpoint atoms and also vanishes. At any other $z \in (0, 1)$, testing the eigenmeasure equation on a shrinking neighborhood of z gives

$$\rho_1 \nu_1(\{z\}) = \frac{1}{2} s(z) \nu_1(\{Tz\}). \quad (9.18)$$

If $\nu_1(\{z\}) > 0$, then $\nu_1(\{Tz\}) \geq 2\rho_1 \nu_1(\{z\})$, and $2\rho_1 > 1$ by (9.3). A periodic forward orbit gives a contradiction after one cycle; a nonperiodic orbit produces infinitely many distinct atoms whose masses grow geometrically, contradicting finiteness. Thus ν_1 , and hence μ , is atomless.

Finally suppose ν_1 had a nonzero absolutely continuous part $r(x) dx$. Weighted composition by Lipschitz branches preserves the Lebesgue decomposition, so this part must itself satisfy the eigenmeasure equation. A change of variables in $\int \mathcal{L}_1 \varphi r dx = \rho_1 \int \varphi r dx$ gives

$$\rho_1 r(x) = s(x) r(Tx) \quad \text{for almost every } x. \quad (9.19)$$

On the set where $r(x) > 0$, iteration yields

$$r(T^n x) = r(x) \frac{\rho_1^n}{\prod_{j=0}^{n-1} s(T^j x)}. \quad (9.20)$$

For Lebesgue-almost every x , Birkhoff's theorem makes the logarithmic growth rate of the right side $\log(2\rho_1) > 0$. On the other hand, for every $\varepsilon > 0$, invariance of Lebesgue measure and Markov's inequality give

$$\sum_{n=1}^{\infty} \text{Leb}\{x : r(T^n x) > e^{\varepsilon n}\} = \sum_{n=1}^{\infty} \text{Leb}\{r > e^{\varepsilon n}\} \leq \|r\|_1 \sum_{n=1}^{\infty} e^{-\varepsilon n} < \infty.$$

Borel–Cantelli implies $r(T^n x) \leq e^{\varepsilon n}$ eventually for almost every x . Choosing $\varepsilon < \log(2\rho_1)$ contradicts (9.20) wherever $r(x) > 0$. Hence the absolutely continuous part is zero and ν_1 is singular. Multiplication by the continuous function f_1 preserves singularity. $\square$

Theorem 9.6 (Exact dyadic phase profile). *For every $y \in [1/2, 1]$ and every fixed $t_0 > 0$,*

$$\lim_{n \rightarrow \infty} \frac{1}{2^n y - t_0} \int_{t_0}^{2^n y} \frac{|\Phi(x)|}{g_1(x)} dx = P(y) := \frac{1 + D(2y - 1)}{2y}. \quad (9.21)$$

The function P is continuous, satisfies $P(1/2) = P(1) = 1$, and is not constant. Therefore the unrestricted ordinary Cesàro limit for the natural gauge g_1 does not exist.

Proof. The spectral decomposition (7.5), first for Hölder test functions and then by upper and lower Hölder approximation of $\mathbf{1}_{[0,z]}$, gives

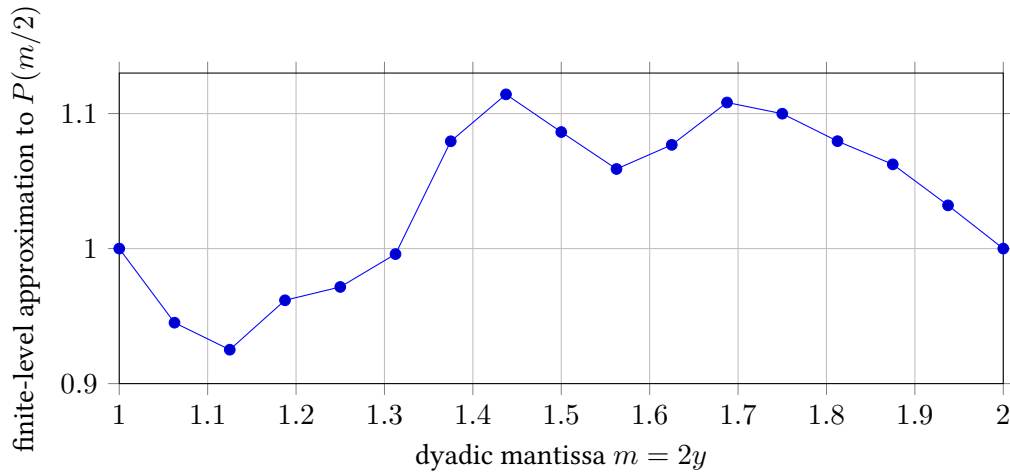
$$\rho_1^{-n} \int_0^z f_1(x) \prod_{j=0}^{n-1} s(T^j x) dx \rightarrow \left(\int_0^1 h_1 \right) \nu_1(f_1 \mathbf{1}_{[0,z]}). \quad (9.22)$$

The squeezing step is valid because ν_1 has no atoms. Returning through $x = 2y - 1$, the partial integral over the current shell satisfies

$$\int_{2^{n-1}}^{2^n y} \frac{|\Phi(x)|}{g_1(x)} dx = 2^{n-1} D(2y - 1) + o(2^n). \quad (9.23)$$

The sum of all preceding complete shells is $2^{n-1} + o(2^n)$ by Theorem 9.4. Dividing the sum of these two contributions by $2^n y$ proves (9.21); a fixed lower endpoint is negligible.

The endpoint values follow from $D(0) = 0$ and $D(1) = 1$. If $P \equiv 1$, then $1 + D(2y - 1) = 2y$, so after putting $z = 2y - 1$ one obtains $D(z) = z$. That would make μ Lebesgue measure, contradicting the singularity in Theorem 9.5. $\square$



What the formula says

Complete shells are normalized correctly, but an additive average gives the last incomplete shell macroscopic weight. Its mass does not become uniform: it converges to the singular conformal measure μ . This is why a power of $\log x$, or any other slowly varying scalar correction, cannot remove the endpoint dependence.

10 How strong is the Cesàro obstruction?

The failure in [Theorem 9.6](#) is not caused merely by choosing the wrong constant. It survives every correction whose variation inside one dyadic shell remains bounded after an arbitrary shellwise rescaling.

On the shell $[2^k, 2^{k+1}]$, use the coordinate $x = 2^k(1+z)$, $0 \leq z \leq 1$, and define the finite measure

$$d\mu_k(z) = D_k(z) dz, \quad D_k(z) = \frac{|\Phi(2^k(1+z))|}{g_1(2^k(1+z))}. \quad (10.1)$$

The transfer-operator limit gives

$$\mu_k \implies \mu, \quad \mu_k([0, 1]) \longrightarrow 1, \quad (10.2)$$

where μ is the singular probability measure in [Theorem 9.5](#).

Theorem 10.1 (Bounded-distortion obstruction). *Let $g : [X_0, \infty) \rightarrow (0, \infty)$ be measurable. Suppose there are constants $0 < m < M < \infty$ and positive shell scalars c_k such that, for all sufficiently large k and every $z \in [0, 1]$,*

$$m \leq \frac{g(2^k(1+z))}{c_k g_1(2^k(1+z))} \leq M. \quad (10.3)$$

Then g cannot satisfy

$$\frac{1}{t - X_0} \int_{X_0}^t \frac{|\Phi(x)|}{g(x)} dx \longrightarrow 1. \quad (10.4)$$

Proof. Assume (10.4). Write its numerator as $I(t)$. For every fixed $0 \leq z_1 < z_2 \leq 1$, subtract the asymptotics at the two endpoints $2^k(1+z_i)$. Since $I(t) = t - X_0 + o(t)$,

$$\frac{I(2^k(1+z_2)) - I(2^k(1+z_1))}{2^k} \longrightarrow z_2 - z_1. \quad (10.5)$$

In particular, after normalization by its total mass, the measure induced by $|\Phi|/g$ on the k th shell must converge weakly to Lebesgue measure on $[0, 1]$.

Remove the irrelevant scalar c_k by putting

$$q_k(z) = \frac{c_k g_1(2^k(1+z))}{g(2^k(1+z))}.$$

Then $M^{-1} \leq q_k \leq m^{-1}$, and the normalized shell probability measure is

$$d\eta_k(z) = \frac{q_k(z) d\mu_k(z)}{\int_0^1 q_k d\mu_k}. \quad (10.6)$$

Let $a_k = \mu_k([0, 1])$ and $\bar{\mu}_k = a_k^{-1} \mu_k$. Because $\int q_k d\mu_k \geq a_k/M$, for every continuous $\varphi \geq 0$,

$$\int \varphi d\eta_k \leq \frac{M}{m} \int \varphi d\bar{\mu}_k. \quad (10.7)$$

Take a weakly convergent subsequence of η_k . Equation (10.2) and (10.7) imply that its limit η is dominated by $(M/m)\mu$. Hence η is singular. But (10.5) forces the entire sequence η_k to converge to Lebesgue measure, a contradiction. $\square$

Corollary 10.2 (Necessary shell microstructure). *Every gauge satisfying (10.4) must have unbounded within-shell distortion relative to g_1 , even after arbitrary scalar rescaling on each shell. In particular, the following corrections cannot work:*

1. *powers or iterated powers of $\log x$;*
2. *regularly varying multipliers;*
3. *fixed continuous log-periodic factors;*
4. *any positive oscillatory multiplier bounded above and bounded away from zero.*

Proof. Each listed class satisfies (10.3) for suitable shell scalars. For a slowly varying multiplier L , the uniform convergence theorem gives $L(2^k(1+z))/L(2^k) \rightarrow 1$ uniformly in $z \in [0, 1]$. Fixed log-periodic or bounded oscillatory multipliers are bounded above and below. The conclusion follows from Theorem 10.1. $\square$

Common pitfall

The theorem does not say that no analytic or no “explicit-looking” gauge can work. It says that a successful gauge must develop increasingly fine and increasingly large relative variation inside successive dyadic shells. That observation points directly to the adaptive construction in the next section.

11 A shell-adaptive analytic gauge with an exact limit

The singular shell measure can be flattened at each finite level. By increasing the resolution slowly enough, the flattening multipliers have logarithms and logarithmic slopes of size $o(k)$ on the k th shell. The baseline $1/g_1$ has slope of order k , so the resulting reciprocal gauge remains increasing, and the gauge itself remains decreasing.

11.1 A quantitative smooth-flattening lemma

Lemma 11.1 (Smooth flattening of weakly convergent measures). *Let σ_k be finite positive Borel measures on $[0, 1]$ such that $\sigma_k \Rightarrow \sigma$ and $\sigma_k([0, 1]) \rightarrow 1$, where σ is an atomless, fully supported probability measure. Then there are positive functions $r_k \in C^\infty[0, 1]$ with the following properties:*

1. *$r_k \equiv 1$ on neighborhoods of both endpoints;*
2. *the weighted cumulative masses become uniformly linear:*

$$\sup_{0 \leq z \leq 1} \left| \int_0^z r_k d\sigma_k - z \right| \rightarrow 0; \quad (11.1)$$

3. *the flattening changes slowly compared with the shell index:*

$$\|\log r_k\|_\infty = o(k), \quad \|(\log r_k)'\|_\infty = o(k). \quad (11.2)$$

Proof. We separate finite-resolution equalization from a diagonal choice of resolution.

Step 1: choose finite partitions. For each $j \geq 1$, choose a partition

$$0 = x_{j,0} < x_{j,1} < \cdots < x_{j,N_j} = 1$$

with mesh at most ε_j , where $\varepsilon_j \downarrow 0$. Because σ is atomless, all boundary points are continuity points. Because σ has full support, every cell $I_{j,i} = [x_{j,i-1}, x_{j,i}]$ has positive σ -mass. For fixed j , weak

convergence gives simultaneous convergence $\sigma_k(I_{j,i}) \rightarrow \sigma(I_{j,i})$ over the finitely many cells. Hence, for large k , the cell equalizers

$$c_{k,j,i} = \frac{|I_{j,i}|}{\sigma_k(I_{j,i})} \quad (11.3)$$

are positive and bounded above and below by constants depending only on j . Choose B_j so that $|\log c_{k,j,i}| \leq B_j/2$ for all these cells and all sufficiently large k .

Step 2: smooth the jumps on a negligible set. Choose a finite union U_j of small intervals around all partition boundaries, including the endpoints, so that $\sigma(\partial U_j) = 0$ and

$$\sigma(U_j) + |U_j| < \frac{\varepsilon_j e^{-B_j}}{20}. \quad (11.4)$$

Weak convergence gives $\sigma_k(U_j) \rightarrow \sigma(U_j)$. Outside U_j , set $\ell_{k,j} = \log c_{k,j,i}$ on the interior of each cell. Inside U_j , join adjacent constant values smoothly, without leaving their convex hull, and arrange $\ell_{k,j} = 0$ near 0 and 1. Since there are only finitely many transitions for fixed j , the interpolation may be chosen with

$$\|\ell_{k,j}\|_\infty \leq B_j, \quad \|\ell'_{k,j}\|_\infty \leq C_j \quad (11.5)$$

for some $C_j < \infty$ independent of sufficiently large k . Put $r_{k,j} = \exp \ell_{k,j}$.

On a complete cell $I = I_{j,i}$, the exact equality $c_{k,j,i} \sigma_k(I) = |I|$ implies

$$\int_I r_{k,j} d\sigma_k - |I| = \int_{I \cap U_j} (r_{k,j} - c_{k,j,i}) d\sigma_k. \quad (11.6)$$

The total error over any union of complete cells is therefore at most $2e^{B_j} \sigma_k(U_j)$. A cumulative interval $[0, z]$ contains at most one partial cell. Away from U_j , the weighted mass of that partial cell is at most its full length, and inside U_j it is at most $e^{B_j} \sigma_k(U_j)$. The missing Lebesgue length is bounded by the mesh and $|U_j|$. By (11.4), for all sufficiently large k ,

$$\sup_z \left| \int_0^z r_{k,j} d\sigma_k - z \right| \leq 3\varepsilon_j. \quad (11.7)$$

Step 3: let the resolution increase diagonally. Choose integers $K_j \uparrow \infty$ so large that all preceding estimates hold for $k \geq K_j$ and, in addition,

$$\frac{B_j}{K_j} \leq \varepsilon_j, \quad \frac{C_j}{K_j} \leq \varepsilon_j. \quad (11.8)$$

For $K_j \leq k < K_{j+1}$, set $r_k = r_{k,j}$. Then $j = j(k) \rightarrow \infty$, so (11.7) proves (11.1); (11.5) and (11.8) prove (11.2). The endpoint condition was built into the smoothing. $\square$

11.2 The smooth adaptive gauge

Apply Theorem 11.1 to $\sigma_k = \mu_k$ and $\sigma = \mu$. On each sufficiently large shell define

$$R(2^k(1+z)) = r_k(z), \quad 0 \leq z \leq 1. \quad (11.9)$$

Because $r_k = 1$ near both endpoints, adjacent shell definitions agree with all derivatives near powers of two. Thus R is C^∞ on a half-line. Put

$$h_0(x) = \frac{1}{g_1(x)}, \quad h(x) = h_0(x)R(x), \quad g_\infty(x) = \frac{1}{h(x)}. \quad (11.10)$$

Lemma 11.2 (The adaptive gauge remains decreasing). *For all sufficiently large x , $h'(x) > 0$ and $g'_\infty(x) < 0$. Moreover,*

$$\log g_\infty(x) = \log g_1(x) + o(\log x). \quad (11.11)$$

Proof. In the shell coordinate $x = 2^k(1+z)$, direct differentiation gives

$$\frac{\partial}{\partial z} \log h_0(2^k(1+z)) = \frac{k + \kappa_1 + a^{-1} \log(1+z)}{1+z}. \quad (11.12)$$

Uniformly in $z \in [0, 1]$, this is at least $k/3$ for large k . On the other hand, $\partial_z \log R = (\log r_k)'$ has supremum norm $o(k)$ by (11.2). Therefore $\partial_z \log h > 0$ eventually. Since z increases with x , $h' > 0$ and $g'_\infty = -(h'/h^2) < 0$.

Also $\log g_\infty = \log g_1 - \log R$. On the k th shell, $\|\log R\|_\infty = o(k)$, while $k \asymp \log x$, proving (11.11). $\square$

Proposition 11.3 (Smooth exact arithmetic gauge). *After an arbitrary positive smooth extension over an initial compact interval, the gauge g_∞ satisfies, for every fixed $X_0 > 0$,*

$$\frac{1}{t - X_0} \int_{X_0}^t \frac{|\Phi(x)|}{g_\infty(x)} dx \longrightarrow 1. \quad (11.13)$$

Proof. Define

$$C_k(z) = \int_0^z D_k(u) r_k(u) du.$$

By Theorem 11.1, $\varepsilon_k := \sup_z |C_k(z) - z| \rightarrow 0$. If $t = 2^K(1+z)$, complete and partial shells give, up to one fixed initial contribution,

$$\int_{X_0}^t \frac{|\Phi(x)|}{g_\infty(x)} dx = \sum_{k < K} 2^k C_k(1) + 2^K C_K(z). \quad (11.14)$$

Replacing $C_k(1)$ by one and $C_K(z)$ by z costs

$$O(1) + \sum_{k < K} 2^k \varepsilon_k + 2^K \varepsilon_K = o(2^K).$$

Indeed, for any $\varepsilon > 0$, all sufficiently late ε_k are below ε , while the finitely many earlier terms are $o(2^K)$. The main term is

$$\sum_{k < K} 2^k + 2^K z = t + O(1),$$

after the fixed initial shells are absorbed into the constant. Division by $t - X_0$ proves (11.13). $\square$

11.3 Real-analytic upgrade

We use Hoischen's derivative form of Carleman approximation: if $f \in C^1(\mathbb{R})$ and $\epsilon : \mathbb{R} \rightarrow (0, \infty)$ is continuous, there is an entire function F , real on $\mathbb{R}$, such that

$$|F(x) - f(x)| < \epsilon(x), \quad |F'(x) - f'(x)| < \epsilon(x) \quad (x \in \mathbb{R}). \quad (11.15)$$

See Gauthier–Kienzle [8] for a convenient modern formulation.

Theorem 11.4 (Positive analytic adaptive gauge). *There exists a positive real-analytic strictly decreasing function $g_{\text{ad}} : (0, \infty) \rightarrow (0, \infty)$ such that, for every fixed $t_0 > 0$,*

$$\frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g_{\text{ad}}(x)} dx \longrightarrow 1, \quad (11.16)$$

and

$$\log g_{\text{ad}}(x) = -\frac{(\log x)^2}{2 \log 2} - \kappa_1 \log x + o(\log x). \quad (11.17)$$

Proof. Extend the function $h = 1/g_\infty$ from a sufficiently far tail to a positive C^1 function on $\mathbb{R}$ with $h'(x) > 0$ everywhere. This can be done on the remaining compact interval by monotone Hermite interpolation, or by joining to a positive exponential. Choose any continuous $\delta(x) > 0$ with $\delta(x) \rightarrow 0$ as $x \rightarrow +\infty$, and then a continuous positive tolerance satisfying

$$\epsilon(x) < \frac{1}{2} \min\{h(x), h'(x), \delta(x)h(x)\}. \quad (11.18)$$

Apply (11.15). The resulting real entire function $\tilde{h}$ obeys $\tilde{h} > h/2 > 0$ and $\tilde{h}' > h'/2 > 0$ on $\mathbb{R}$, and

$$\frac{\tilde{h}(x)}{h(x)} \rightarrow 1. \quad (11.19)$$

Define $g_{\text{ad}} = 1/\tilde{h}$ on $(0, \infty)$. It is positive, real analytic, and strictly decreasing.

Equation (11.19), together with (11.11), proves the logarithmic scale (11.17). To preserve the average, fix $\varepsilon > 0$ and choose X so that $|\tilde{h}/h - 1| < \varepsilon$ for $x \geq X$. Then

$$\begin{aligned} \left| \int_X^t |\Phi(x)| (\tilde{h}(x) - h(x)) dx \right| &\leq \varepsilon \int_X^t |\Phi(x)| h(x) dx \\ &= \varepsilon(t + o(t)) \end{aligned}$$

by Theorem 11.3. The compact initial interval contributes $O(1)$. Divide by t , let $t \rightarrow \infty$, and then let $\varepsilon \downarrow 0$. This proves (11.16). $\square$

Remark 11.5. There is no contradiction with Theorem 10.1. At level k , the flattening corrects only a finite partition, but the number of cells tends to infinity. Its logarithm and logarithmic slope are $o(k)$, so monotonicity survives, while its distortion relative to g_1 becomes unbounded and its shell shape never stabilizes.

12 The multiplicatively natural logarithmic mean

The endpoint phase is specific to additive averaging. Under dx/x , every complete dyadic shell has the same weight $\log 2$, and one incomplete shell is negligible among N completed shells.

Theorem 12.1 (Unrestricted logarithmic mean). *For every fixed $t_0 > 1$,*

$$\frac{1}{\log(t/t_0)} \int_{t_0}^t \frac{|\Phi(x)|}{E_{\kappa_1}(x)} \frac{dx}{x} \rightarrow A_1^{\log}, \quad (12.1)$$

where

$$A_1^{\log} = \frac{1}{\log 2} \left(\int_0^1 h_1(x) dx \right) \nu_1 \left(\frac{f_1}{1 + \cdot} \right) > 0. \quad (12.2)$$

Numerically,

$$A_1^{\log} = 0.09386063575678126354 \dots \quad (12.3)$$

Proof. For the shell $[2^{n-1}, 2^n]$, put

$$\beta_n = \int_{2^{n-1}}^{2^n} \frac{|\Phi(x)|}{E_{\kappa_1}(x)} \frac{dx}{x}.$$

Use $x = 2^n y$, then $z = 2y - 1$. Since $dx/x = dz/(1+z)$, the master factorization gives

$$\beta_n = \rho_1^{-n} \int_0^1 \frac{f_1(z)}{1+z} \prod_{j=0}^{n-1} s(T^j z) dz. \quad (12.4)$$

The RPF limit implies

$$\beta_n \longrightarrow B_1 := \left(\int_0^1 h_1 \right) \nu_1 \left(\frac{f_1}{1 + \cdot} \right).$$

If $2^N \leq t < 2^{N+1}$, the logarithmic numerator is the sum of $N + O(1)$ complete-shell terms β_n , plus one partial-shell term. The latter is uniformly bounded because it is bounded by the full positive shell integral. Hence the numerator is $NB_1 + o(N)$, while $\log(t/t_0) = Na + O(1)$. Their quotient tends to B_1/a , which is (12.2). $\square$

13 The sharp uniform envelope

The preceding norms describe the bulk of a dyadic shell. We now ask for the largest possible numerator. This is an ergodic-optimization problem, but in the dyadic case its sharp exponential rate follows from one elementary two-step inequality.

Put

$$\lambda := \frac{\sqrt{3}}{2}, \quad \beta := \log \lambda < 0. \quad (13.1)$$

Lemma 13.1 (Sharp two-step sine inequality). *For every real x ,*

$$\frac{2}{3} \log s(x) + \frac{1}{3} \log s(Tx) \leq \log \lambda, \quad (13.2)$$

where the left-hand side is interpreted as $-\infty$ if one of the factors vanishes. Equality holds precisely when $x \equiv 1/3$ or $2/3 \pmod{1}$.

Proof. Since all factors are nonnegative, exponentiating three times shows that (13.2) is equivalent to

$$s(x)^2 s(Tx) \leq \lambda^3. \quad (13.3)$$

Let $\theta = \pi x$ and $u = \sin^2 \theta$. Because $s(Tx) = |\sin(2\theta)| = 2|\sin \theta \cos \theta|$,

$$s(x)^2 s(Tx) = 2|\sin^3 \theta \cos \theta|.$$

Squaring reduces the maximization to

$$4u^3(1-u), \quad 0 \leq u \leq 1.$$

Its derivative is $4u^2(3-4u)$; hence its unique interior maximum is attained at $u = 3/4$, and the maximum value is $27/64$. Taking square roots gives

$$s(x)^2 s(Tx) \leq \frac{3\sqrt{3}}{8} = \left(\frac{\sqrt{3}}{2} \right)^3 = \lambda^3.$$

Equality means $\sin^2(\pi x) = 3/4$, which is equivalent to $x \equiv 1/3$ or $2/3 \pmod{1}$. $\square$

For later use define

$$Q_n(x) := \prod_{j=0}^{n-1} s(T^j x), \quad n \geq 1. \quad (13.4)$$

Proposition 13.2 (Sharp exponential growth of the lacunary product). *For every $n \geq 1$,*

$$\lambda^n \leq \|Q_n\|_\infty \leq \lambda^{n-1}. \quad (13.5)$$

Consequently

$$\lim_{n \rightarrow \infty} \|Q_n\|_\infty^{1/n} = \lambda. \quad (13.6)$$

Proof. At $x = 1/3$, the doubling orbit alternates between $1/3$ and $2/3$, and every factor equals λ . Thus $Q_n(1/3) = \lambda^n$, proving the lower bound.

For the upper bound put $\phi = \log s$. For $n \geq 2$, apply [Theorem 13.1](#) to $x, Tx, \dots, T^{n-2}x$ and add the resulting inequalities:

$$\frac{2}{3}\phi(x) + \sum_{j=1}^{n-2} \phi(T^j x) + \frac{1}{3}\phi(T^{n-1}x) \leq (n-1) \log \lambda.$$

Now add $\frac{1}{3}\phi(x) + \frac{2}{3}\phi(T^{n-1}x) \leq 0$. We obtain

$$\sum_{j=0}^{n-1} \phi(T^j x) \leq (n-1) \log \lambda.$$

Exponentiation gives $Q_n(x) \leq \lambda^{n-1}$. The case $n = 1$ is simply $s(x) \leq 1 = \lambda^0$. Taking n th roots in the two-sided estimate proves (13.6). $\square$

The extremizing orbit is the period-two orbit $1/3 \leftrightarrow 2/3$. In terms of the Thue–Morse polynomial, (13.6) is the classical Gelfond exponent; see [9, 7].

Theorem 13.3 (Sharp global Fourier envelope). *Define*

$$\kappa_\infty := \frac{\log \pi + a/2 - \log \lambda}{a} = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2} = 2.3590148791117407073\dots \quad (13.7)$$

Then there is a finite constant C such that

$$|\Phi(x)| \leq CE_{\kappa_\infty}(x), \quad x \geq 1. \quad (13.8)$$

The exponent is sharp: for every $n \geq 0$,

$$|\Phi(2^n \cdot 2/3)| = C_* E_{\kappa_\infty}(2^n \cdot 2/3), \quad C_* := W_{\kappa_\infty}(2/3) > 0. \quad (13.9)$$

Proof. The definition of κ_∞ is precisely the cancellation identity

$$\exp(n(\kappa_\infty a - \log \pi - a/2)) = \lambda^{-n}. \quad (13.10)$$

Use the normalized master identity (3.4). Its lacunary factor is a translate of a product of the form Q_n , so [Theorem 13.2](#) gives

$$\frac{|\Phi(2^n y)|}{E_{\kappa_\infty}(2^n y)} \leq \lambda^{-1} \max_{1/2 \leq y \leq 1} W_{\kappa_\infty}(y).$$

After increasing the harmless constant to cover $n = 0$, every $x \geq 1$ has a unique representation $x = 2^n y$ with $y \in [1/2, 1)$, proving (13.8).

For $y = 2/3$, the orbit alternates between $2/3$ and $1/3$, hence every sine factor in (3.4) equals λ . The factor λ^n cancels (13.10) exactly, leaving the constant $W_{\kappa_\infty}(2/3)$. This proves (13.9) and sharpness. $\square$

Corollary 13.4 (Sharp log-square order). *With $\log 0 = -\infty$,*

$$\limsup_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\frac{1}{2 \log 2}, \quad \liminf_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\infty. \quad (13.11)$$

In particular, $\Phi(x) = O(x^{-M})$ for every $M > 0$, but $e^{cx} |\Phi(x)|$ is unbounded for every $c > 0$.

Proof. The envelope gives the upper bound for the limsup, and the exact ray (13.9) gives the reverse inequality. The liminf is $-\infty$ because Φ vanishes at every positive integer. The negative quadratic in $\log x$ dominates every linear term $-M \log x$, proving super-polynomial decay. Along the ray in (13.9), however, the logarithm is only $-O((\log x)^2)$, which is asymptotically much larger than $-cx$. $\square$

What the formula says

The universal log-square coefficient is a genuine pointwise fact, but the power $x^{-\kappa}$ that follows it still depends on how values are selected. The exponent κ_∞ describes the best possible points, not a typical point and not every local maximum.

14 From lobe height to lobe area

For each integer $m \geq 1$, let

$$M_m := \max_{m \leq x \leq m+1} |\Phi(x)| = |\Phi(\xi_m)|, \quad (14.1)$$

where ξ_m is the unique maximizer supplied by [Theorem 2.4](#). Log-concavity implies more than uniqueness: the height and the area of a lobe differ by at most a polylogarithmic factor.

Lemma 14.1 (Peak–area comparison). *There is an absolute constant C such that, for every integer $m \geq 2$,*

$$\int_m^{m+1} |\Phi(x)| \, dx \leq M_m \leq C(1 + \log m)^{5/2} \int_m^{m+1} |\Phi(x)| \, dx. \quad (14.2)$$

Proof. The first inequality is immediate because the interval has length one. We prove the second quantitatively. On the lobe $(m, m+1)$, write

$$\ell(x) = \log |\Phi(x)|, \quad d_0 = 1 + \nu_2(m), \quad d_1 = 1 + \nu_2(m+1),$$

and set $\delta_0 = \xi_m - m$, $\delta_1 = m+1 - \xi_m$.

From the logarithmic derivative formula (2.7), isolate the terms corresponding to the adjacent zeros m and $m+1$. For $m < x < m+1$,

$$\ell'(x) = \frac{d_0 \theta_0(x)}{x - m} - \frac{d_1 \theta_1(x)}{m+1 - x} + R_m(x), \quad (14.3)$$

where

$$\theta_0(x) = \frac{2x}{x+m}, \quad \theta_1(x) = \frac{2x}{x+m+1}.$$

For $m \geq 2$, both θ_i lie between $4/5$ and $6/5$. We claim that

$$|R_m(x)| \leq C_0(1 + \log m)^2 \quad (m < x < m+1). \quad (14.4)$$

Indeed, for $k \leq 2m$, excluding $m, m+1$, the k th term in (2.7) is bounded by

$$C \frac{1 + \nu_2(k)}{|k - x|} \leq C \frac{1 + \log m}{|k - x|}.$$

Summing the reciprocal distances gives $O(\log m)$. For $k > 2m$, the term is at most $Cm(1 + \log k)/k^2$, whose tail is $O(1 + \log m)$. This proves (14.4).

At $x = \xi_m$, equation $\ell'(\xi_m) = 0$ and (14.3) force the maximizer away from both endpoints. For example, if $\delta_0 \leq 1/2$, then $\delta_1 \geq 1/2$, so

$$\frac{4d_0}{5\delta_0} \leq \frac{6d_1}{5\delta_1} + |R_m(\xi_m)| \leq C(1 + \log m)^2.$$

Since $d_0 \geq 1$, $\delta_0 \geq c(1 + \log m)^{-2}$. Interchanging the endpoints gives

$$\delta_* := \min(\delta_0, \delta_1) \geq c(1 + \log m)^{-2}. \quad (14.5)$$

Differentiate once more. The exact formula (2.8) shows that on $|x - \xi_m| \leq \delta_*/2$, the two adjacent terms contribute at most $C(1 + \log m)\delta_*^{-2}$, while the remaining inverse-square sum is smaller. By (14.5),

$$|\ell''(x)| \leq \Lambda_m := C_1(1 + \log m)^5. \quad (14.6)$$

Taylor's formula at the critical point, together with $\ell'(\xi_m) = 0$, gives

$$\ell(\xi_m + u) \geq \ell(\xi_m) - \frac{1}{2}\Lambda_m u^2.$$

Choose

$$w_m = \min\{\delta_*/2, \Lambda_m^{-1/2}\} \geq c_1(1 + \log m)^{-5/2}.$$

For $|u| \leq w_m$, therefore, $|\Phi(\xi_m + u)| \geq e^{-1/2}M_m$. Integrating on this interval yields

$$\int_m^{m+1} |\Phi(x)| \, dx \geq 2e^{-1/2}w_m M_m,$$

which rearranges to the second inequality in (14.2). $\square$

Remark 14.2. The exponent $5/2$ is a robust worst-case bound, not a prediction of the actual ratio. Numerical data suggest much tighter behavior, but no limiting peak-to-area constant is used in this article.

15 The largest value in a dyadic annulus

The global envelope in Theorem 13.3 compares a value at x with the scale $E_{\kappa_\infty}(x)$ at that same point. A subtler problem is

$$\mathfrak{M}_k := \max_{2^k \leq x \leq 2^{k+1}} |\Phi(x)|. \quad (15.1)$$

If the whole annulus is compared with the scale at its left endpoint 2^k , the maximizer moves toward that endpoint. It cannot reach the endpoint, because 2^k is a zero. The competition between these two effects creates a Lambert- W boundary layer.

15.1 Exact reduction to a weighted lacunary product

Let $n = k + 1$, and define the bounded tail

$$\mathcal{H}(y) := \prod_{m=1}^{\infty} \left| \operatorname{sinc}\left(\frac{\pi y}{2^m}\right) \right| = |\Phi(y/2)|. \quad (15.2)$$

Write $x = 2^k(1 + \theta)$, with $0 \leq \theta \leq 1$. Splitting the sinc product after its first n factors gives the exact identity

$$\left| \Phi(2^k(1 + \theta)) \right| = \frac{2^{-k(k+1)/2}}{\pi^n} \mathcal{H}(1 + \theta)(1 + \theta)^{-n} Q_n(\theta). \quad (15.3)$$

Indeed, for $0 \leq h \leq k$, the absolute value of the sine numerator is $|\sin(\pi 2^{k-h}\theta)|$. Reversing the order produces $Q_n(\theta)$, while

$$\prod_{h=0}^k \frac{\pi 2^k(1 + \theta)}{2^h} = \pi^n (1 + \theta)^n 2^{k(k+1)/2}.$$

The remaining factors are exactly $\mathcal{H}(1 + \theta)$.

Thus

$$\mathfrak{M}_k = \pi^{-n} 2^{-k(k+1)/2} A_n, \quad A_n := \sup_{0 \leq \theta \leq 1} \mathcal{H}(1 + \theta)(1 + \theta)^{-n} Q_n(\theta). \quad (15.4)$$

The factor $(1 + \theta)^{-n}$ favors $\theta \approx 0$, whereas $Q_n(0) = 0$.

15.2 Transition coordinates near the zero

For $0 < \theta \leq 1/2$, write

$$\theta = 2^{-r}s, \quad r \in \{0, 1, 2, \dots\}, \quad s \in I := [1/4, 1/2]. \quad (15.5)$$

Apart from harmless endpoint duplications, this representation is unique. If $r < n$, then

$$Q_n(2^{-r}s) = R_r(s)Q_{n-r}(s), \quad R_r(s) := \prod_{h=1}^r \sin\left(\frac{\pi s}{2^h}\right). \quad (15.6)$$

The first block is uniformly explicit:

Lemma 15.1 (Small-angle prefix). *Uniformly for $s \in I$,*

$$\log R_r(s) = -\frac{a}{2}r^2 + r \left(\log(\pi s) - \frac{a}{2} \right) + O(1) \quad (r \rightarrow \infty). \quad (15.7)$$

Proof. Write every factor as its argument times a sinc:

$$\begin{aligned} \log R_r(s) &= \sum_{h=1}^r \left[\log(\pi s) - ha + \log \operatorname{sinc}\left(\frac{\pi s}{2^h}\right) \right] \\ &= r \log(\pi s) - \frac{a}{2}r(r+1) + \sum_{h=1}^r \log \operatorname{sinc}\left(\frac{\pi s}{2^h}\right). \end{aligned}$$

Since $s \in [1/4, 1/2]$, the last series converges uniformly as $r \rightarrow \infty$: its terms are $O(4^{-h})$, uniformly in s . Rearranging the quadratic and linear terms proves (15.7). $\square$

The second block can attain the Gelfond rate after a short transient, uniformly near any prescribed s .

Lemma 15.2 (Nearby preimages of the maximizing cycle). *There is an absolute constant C such that, for every $s_0 \in I$ and every integer $q \geq 1$, one can find $s_q \in I$ satisfying*

$$|s_q - s_0| \leq 2^{-q}, \quad T^q s_q \in \{1/3, 2/3\}, \quad (15.8)$$

and, for every $m \geq q$,

$$Q_m(s_q) \geq \exp(m\beta - Cq^2). \quad (15.9)$$

Proof. The q -fold preimages

$$\frac{j+1/3}{2^q}, \quad \frac{j+2/3}{2^q} \quad (j \in \mathbb{Z})$$

form two grids of mesh 2^{-q} . One of their points lies in I within distance 2^{-q} of s_0 , proving (15.8).

For $0 \leq j < q$, the number $T^j s_q$ is a nonintegral rational whose reduced denominator is at most $3 \cdot 2^{q-j}$. Hence its distance from the nearest integer is at least $(3 \cdot 2^{q-j})^{-1}$. Since $|\sin(\pi u)| \geq 2 \operatorname{dist}(u, \mathbb{Z})$,

$$s(T^j s_q) \geq \frac{2}{3 \cdot 2^{q-j}}.$$

The logarithmic loss in the first q factors is therefore at most Cq^2 . After time q , the orbit alternates between $1/3$ and $2/3$, so every later factor equals $\lambda = e^\beta$. Multiplying the transient and periodic parts gives (15.9). $\square$

15.3 The finite-dimensional boundary-layer objective

Define

$$F_n(r, s) := -\frac{a}{2}r^2 + r \left(\log(\pi s) - \frac{a}{2} - \beta \right) - n \log(1 + s2^{-r}). \quad (15.10)$$

Lemma 15.3 (Reduction to F_n). *As $n \rightarrow \infty$,*

$$\log A_n = n\beta + \sup_{\substack{r \geq 0 \text{ integer} \\ s \in I}} F_n(r, s) + O((\log \log n)^2). \quad (15.11)$$

Moreover, every pair that is within $O((\log \log n)^2)$ of the supremum satisfies

$$r = O(\log n), \quad ns2^{-r} = O((\log n)^2). \quad (15.12)$$

Proof. Suppose first that $0 < \theta \leq 1/2$, write $\theta = 2^{-r}s$, and assume $r < n$. By [Theorems 13.2](#) and [15.1](#),

$$\log Q_n(\theta) \leq -\frac{a}{2}r^2 + r \left(\log(\pi s) - \frac{a}{2} \right) + (n-r)\beta + O(1).$$

Since $\mathcal{H}$ is bounded on $[1, 2]$, this yields

$$\log [\mathcal{H}(1+\theta)(1+\theta)^{-n}Q_n(\theta)] \leq n\beta + F_n(r, s) + O(1). \quad (15.13)$$

The omitted regions cannot contain a near-maximizer. If $\theta \geq 1/2$, the weight alone contributes at most $-(\log(3/2))n$, which is exponentially below the trial value described below. If $r \geq n$, then all n sine arguments are small and $|\sin u| \leq |u|$ gives a logarithm of order $-\Omega(n^2)$. On the other hand, take $r = \lfloor \log_2 n \rfloor$ and $s = 1/3$. The tail lies exactly on the maximizing two-cycle, while the prefix loses only $O((\log n)^2)$. Thus

$$\log A_n \geq n\beta - O((\log n)^2). \quad (15.14)$$

This proves that a near-maximizer lies in the represented range $r < n$.

Because $s \in I$,

$$F_n(r, s) \leq -\frac{a}{2}r^2 + C_1r.$$

Comparison with [\(15.14\)](#) first gives $r = O(\log n)$. Also $s2^{-r} \leq 1/2$, so $\log(1+u) \geq u/2$ gives

$$F_n(r, s) \leq -\frac{a}{2}r^2 + C_1r - \frac{1}{2}ns2^{-r}.$$

A second comparison with the trial value proves the localization [\(15.12\)](#). Taking the supremum in [\(15.13\)](#) now proves the required upper bound.

For the lower bound, choose a localized pair (r, s) within one of the supremum and put

$$q = \lceil 4 \log_2 \log(n + e^e) \rceil.$$

For large n , $n - r \geq q$. Choose s_q by [Theorem 15.2](#). On the localized region,

$$\left| \frac{\partial F_n}{\partial s} \right| \leq 4r + n2^{-r} = O((\log n)^2).$$

Because $|s_q - s| \leq 2^{-q} = O((\log n)^{-4})$, replacing s by s_q changes F_n by $O(1)$. The dense-preimage lemma supplies

$$\log Q_{n-r}(s_q) \geq (n-r)\beta - Cq^2,$$

while the small-angle prefix gives the matching lower estimate for $R_r(s_q)$. The localization forces $r \rightarrow \infty$, so $\mathcal{H}(1 + 2^{-r}s_q) \rightarrow \mathcal{H}(1) > 0$. Hence the value at this phase is at least

$$n\beta + \sup F_n - O(q^2).$$

Since $q^2 = O((\log \log n)^2)$, the lower bound completes [\(15.11\)](#). □

15.4 Lambert- W optimization

Set

$$C_0 := \log \pi - \frac{a}{2} - \beta = \log \left(\pi \sqrt{\frac{2}{3}} \right), \quad c := ae^{-C_0} = \frac{a}{\pi} \sqrt{\frac{3}{2}}. \quad (15.15)$$

Within the localized range (15.12),

$$n |\log(1 + s2^{-r}) - s2^{-r}| \leq Cn(s2^{-r})^2 = O\left(\frac{(\log n)^4}{n}\right) = o(1).$$

Thus, up to $o(1)$, the objective is

$$-\frac{a}{2}r^2 + r(C_0 + \log s) - ns2^{-r}. \quad (15.16)$$

Put

$$A = C_0 + \log s, \quad z = ar - A.$$

A direct completion of the square gives

$$-\frac{a}{2}r^2 + rA - ns2^{-r} = -\frac{z^2}{2a} - ne^{-C_0}e^{-z} + \frac{A^2}{2a}. \quad (15.17)$$

As s runs through $[1/4, 1/2]$, the interval of possible A 's has length exactly a . As the integer r changes by one, the corresponding z -interval shifts by exactly a . These intervals therefore tile a half-line. Since A remains bounded, the supremum differs by only $O(1)$ from the unrestricted maximum of

$$\varphi_n(z) = -\frac{z^2}{2a} - ne^{-C_0}e^{-z}. \quad (15.18)$$

This function is strictly concave. Its critical equation is

$$-\frac{z}{a} + ne^{-C_0}e^{-z} = 0 \quad \Longleftrightarrow \quad ze^z = cn.$$

Hence the maximizer is

$$w_n = W(cn), \quad (15.19)$$

where W is the principal real Lambert function [3], and

$$\max_z \varphi_n(z) = -\frac{w_n^2 + 2w_n}{2a}. \quad (15.20)$$

Theorem 15.4 (Sharp dyadic-annulus maximum). *Let $w_k = W(c(k+1))$. Then*

$$\begin{aligned} \log \mathfrak{M}_k = & -\frac{a}{2}k(k+1) - (k+1)\log \pi + (k+1)\beta \\ & - \frac{w_k^2 + 2w_k}{2a} + O((\log \log k)^2), \end{aligned} \quad (15.21)$$

Equivalently,

$$\log \mathfrak{M}_k = -\frac{a}{2}k^2 - a\kappa_\infty k - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2a} + O((\log \log k)^2). \quad (15.22)$$

If $2^k(1 + \theta_k)$ is a maximizing point, then

$$\theta_k = \frac{W(c(k+1))}{a(k+1)}e^{o(1)}. \quad (15.23)$$

Proof. Combine the exact relation (15.4), the reduction (15.11), and the maximum (15.20). This gives (15.21). Since $a\kappa_\infty = a/2 + \log \pi - \beta$, collecting the quadratic and linear terms in k proves (15.22); a bounded constant is absorbed by the stated error.

It remains to locate the maximizer. Let $z = w_n + u$. At the optimum, $ne^{-C_0-w_n} = w_n/a$, and therefore

$$\varphi'_n(w_n + u) = \frac{w_n(e^{-u} - 1) - u}{a}.$$

For small u , the loss from the maximum is bounded below by a constant multiple of $w_n u^2$; for $|u| \geq 1$, it is at least a constant multiple of w_n . Because the total uncertainty in Theorem 15.3 is $O((\log \log n)^2) = o(w_n)$, every true maximizer has

$$z = w_n + o(1).$$

Finally,

$$\theta = s2^{-r} = e^{-C_0-z} = e^{-C_0-w_n+o(1)}.$$

The relation $w_n e^{w_n} = cn = a n e^{-C_0}$ gives $e^{-C_0-w_n} = w_n/(an)$. Recalling $n = k + 1$ proves (15.23). $\square$

To express the result directly in terms of the left endpoint $R = 2^k$, put $L = \log R$ and $d = \pi^{-1}\sqrt{3}/2$. Since $c(k + 1) = d(L + a)$,

$$\begin{aligned} \log \max_{R \leq x \leq 2R} |\Phi(x)| &= -\frac{L^2}{2a} - \kappa_\infty L \\ &\quad - \frac{W(d(L + a))^2 + 2W(d(L + a))}{2a} + O((\log \log \log R)^2). \end{aligned} \quad (15.24)$$

The standard expansion of W gives the first two orders of the additional loss:

$$-\frac{(\log \log R)^2}{2a} + \frac{\log \log R \log \log \log R}{a} + O(\log \log R). \quad (15.25)$$

What the formula says

The Lambert term is not a fifth member of the L^q spectrum. It appears because the reference scale is frozen at the left endpoint of a moving annulus. The maximizing point slides toward the zero at that endpoint, at distance about $R \log \log R / \log R$, and pays an additional log-normal penalty.

16 Exceptional valleys near dyadic boundaries

The sharp envelope is attained on special orbits, but many individual lobes are much smaller. The most transparent exceptional family lies immediately after a large power of two. The following exact identity is the basic tool.

Lemma 16.1 (Dyadic-boundary identity). *Let $N = 2^{n-1}$ with $n \geq 1$, and let $0 < z < N$. Then*

$$|\Phi(N + z)| = \frac{|\Phi(\frac{1}{2} + z/(2N))|}{|\Phi(z/(2N))|} \left(\frac{z}{N + z} \right)^n |\Phi(z)|. \quad (16.1)$$

Proof. Compare the products defining $\Phi(N + z)$ and $\Phi(z)$. For $0 \leq h < n$, the number $N/2^h$ is an integer, so

$$\left| \sin(\pi(N + z)/2^h) \right| = \left| \sin(\pi z/2^h) \right|.$$

The quotient of the first n sinc factors is therefore $(z/(N+z))^n$. Reindex the tails with $h = n + j$. Since $2N = 2^n$,

$$\prod_{h=n}^{\infty} \operatorname{sinc}\left(\frac{\pi(N+z)}{2^h}\right) = \Phi\left(\frac{1}{2} + \frac{z}{2N}\right),$$

$$\prod_{h=n}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^h}\right) = \Phi\left(\frac{z}{2N}\right).$$

The denominator on the right is nonzero because $0 < z/(2N) < 1/2$. Taking absolute values and combining the finite and infinite parts proves (16.1). $\square$

16.1 The first lobe after an odd multiple of a power of two

Theorem 16.2 (First-lobe valley after $m2^k$). *Fix an odd positive integer m . Write the unique maximizer in $(m2^k, m2^k + 1)$ as*

$$m2^k + t_k, \quad 0 < t_k < 1.$$

Then, as $k \rightarrow \infty$,

$$(k+1)(1-t_k) \rightarrow 1, \quad (16.2)$$

and

$$M_{m2^k} \sim \frac{\mathcal{H}(1)\mathcal{H}(m)}{e(k+1)} (m2^k)^{-(k+1)}. \quad (16.3)$$

In particular,

$$M_{2^k} \sim \frac{\mathcal{H}(1)^2}{e(k+1)} 2^{-k(k+1)}. \quad (16.4)$$

Proof. Put $N = m2^k$ and $p = k+1$. Splitting the product after its first p factors and using the integral periodicity of the sine numerators gives the exact identity

$$|\Phi(N+t)| = \left(\frac{t}{N+t}\right)^p \frac{|\Phi(t)|}{|\Phi(t/2^p)|} \mathcal{H}\left(m + \frac{t}{2^k}\right), \quad 0 \leq t \leq 1. \quad (16.5)$$

To verify it, note that for $0 \leq h \leq k$, $N/2^h = m2^{k-h}$ is an integer. Hence the first p sine numerators have the same absolute values as those in $\Phi(t)$, and the denominators contribute $(t/(N+t))^p$. The quotient of the remaining tails is exactly the last two factors on the right of (16.5).

Uniformly for $0 \leq t \leq 1$,

$$\Phi(t/2^p) \rightarrow 1, \quad \mathcal{H}(m+t/2^k) \rightarrow \mathcal{H}(m), \quad (1+t/N)^{-p} \rightarrow 1.$$

The last convergence follows from $p/N \rightarrow 0$. Consequently, with a positive multiplier that tends uniformly to one,

$$|\Phi(N+t)| = N^{-p} \mathcal{H}(m) t^p \Phi(t) (1+o(1)). \quad (16.6)$$

Here $\Phi(t) > 0$ on $[0, 1)$.

By Theorem 2.3, the zero at 1 is simple and

$$\Phi(t) = \mathcal{H}(1)(1-t)(1+O(1-t)) \quad (t \uparrow 1). \quad (16.7)$$

We first localize the maximum. Fix $\delta > 0$. On $[0, 1-\delta]$, the factor t^p is at most $(1-\delta)^p$, whereas at $t = 1 - 1/p$, (16.7) gives a value of order p^{-1} . Thus the maximizer tends to 1.

On a fixed neighborhood of 1, the logarithmic derivative of the multiplier omitted in (16.6) is $O(p/N) + O(2^{-k}) = o(1)$. Therefore the exact critical-point equation is

$$\frac{p}{t_k} + \frac{\Phi'(t_k)}{\Phi(t_k)} + o(1) = 0. \quad (16.8)$$

Equation (16.7) gives

$$\frac{\Phi'(t)}{\Phi(t)} = -\frac{1}{1-t} + O(1).$$

Substitution into (16.8), together with $t_k \rightarrow 1$, yields $p(1 - t_k) \rightarrow 1$, proving (16.2). It follows that

$$t_k^p \rightarrow e^{-1}, \quad p\Phi(t_k) \rightarrow \mathcal{H}(1).$$

Insert these limits into (16.6) to obtain (16.3). The case $m = 1$ is (16.4). $\square$

16.2 Every fixed offset after a power of two

The preceding theorem studies the first lobe. The same mechanism works at every fixed offset, with the order of the terminal zero determining the polynomial correction.

Theorem 16.3 (Fixed-offset dyadic valleys). *Fix an integer $r \geq 0$, and put*

$$b = r + 1, \quad d = 1 + \nu_2(b), \quad N = 2^{n-1}. \quad (16.9)$$

Let ξ_{N+r} be the unique maximizer in $(N + r, N + r + 1)$, and put $M_{N+r} = |\Phi(\xi_{N+r})|$. Then

$$\xi_{N+r} = N + b - \frac{db}{n} + o(n^{-1}), \quad (16.10)$$

$$M_{N+r} \sim \left| \Phi(1/2)\Phi(b/2^d) \right| \left(\frac{b}{N} \right)^n \left(\frac{d}{en} \right)^d. \quad (16.11)$$

Consequently

$$\log M_{N+r} = -\frac{(\log N)^2}{\log 2} + O(\log N). \quad (16.12)$$

The leading log-square coefficient is twice the generic coefficient in E_κ .

Proof. Write points in the lobe as $x = N + z$, where $r < z < b$. In (16.1), separate the power N^{-n} :

$$|\Phi(N + z)| = N^{-n} A_n(z), \quad A_n(z) = G_n(z) z^n |\Phi(z)|, \quad (16.13)$$

where

$$G_n(z) = \frac{|\Phi(\frac{1}{2} + z/(2N))|}{|\Phi(z/(2N))|} \left(1 + \frac{z}{N} \right)^{-n}. \quad (16.14)$$

Uniformly for $z \in [r, b]$,

$$G_n(z) \longrightarrow G_\infty := |\Phi(1/2)|. \quad (16.15)$$

Indeed, the two arguments in the quotient converge uniformly to $1/2$ and 0 , while $n/N \rightarrow 0$. For large n , the denominator is nonzero on the whole interval. The continuous extension of A_n vanishes at both endpoints: at $z = b$ because $\Phi(b) = 0$, and at $z = r$ either because $r \geq 1$ is an integer zero or, when $r = 0$, because of the factor z^n . Therefore the exact lobe maximizer is also the exact maximizer of A_n on $[r, b]$.

By the local zero formula (2.5),

$$|\Phi(z)| = D_b(b - z)^d(1 + O(b - z)), \quad D_b = b^{-d} \left| \Phi(b/2^d) \right|. \quad (16.16)$$

Choose a fixed $\delta_0 > 0$ so that this expansion bounds $|\Phi(z)|$ above and below by positive multiples of $(b - z)^d$ on $[b - \delta_0, b]$. The convergence (16.15) gives similar uniform bounds for G_n . At the trial point $z = b - db/n$,

$$A_n(z) \geq c b^{n+d} n^{-d}.$$

On the other hand, when $z \leq b - \delta_0$, the factor z^n is exponentially smaller than b^n . Hence every exact maximizer lies in $[b - \delta_0, b]$ for large n .

Introduce the boundary coordinate

$$u = \frac{n(b - z)}{b}, \quad z = b \left(1 - \frac{u}{n}\right). \quad (16.17)$$

Using $(1 - u/n)^n \leq e^{-u}$ and the local upper bound, we get

$$A_n(z) \leq C b^{n+d} n^{-d} u^d e^{-u}.$$

Because $u^d e^{-u} \rightarrow 0$ as $u \rightarrow \infty$, comparison with the trial point shows that the maximizing u -coordinates remain in one fixed compact interval $[0, U]$. On that compact interval, uniformly in u , equations (16.15) and (16.16) yield

$$\begin{aligned} & \frac{A_n(b - bu/n)}{G_\infty D_b b^{n+d} n^{-d}} \\ &= \frac{G_n(b - bu/n)}{G_\infty} u^d \left(1 - \frac{u}{n}\right)^n (1 + o(1)) \rightarrow u^d e^{-u}. \end{aligned} \quad (16.18)$$

The limiting function has a unique maximum at $u = d$, of value $d^d e^{-d}$. Uniform convergence on $[0, U]$ implies that the exact maximizing coordinates satisfy $u_n \rightarrow d$, and that their maximum values converge to the maximum of the limiting profile. Translating back gives

$$b - (\xi_{N+r} - N) = \frac{db}{n} + o(n^{-1}),$$

which is (16.10), and

$$\max A_n \sim G_\infty D_b b^{n+d} n^{-d} d^d e^{-d}.$$

Multiply by N^{-n} , substitute $D_b = b^{-d} |\Phi(b/2^d)|$, and obtain (16.11).

Finally, $N = 2^{n-1}$, so

$$-n \log N = -\frac{(\log N)^2}{a} + O(\log N).$$

All remaining factors in (16.11) contribute only $O(\log N)$, proving (16.12). $\square$

For the first interval after $N = 2^{n-1}$, one has $r = 0$, $b = d = 1$, and

$$M_N \sim \frac{\Phi(1/2)^2}{en} N^{-n}, \quad \xi_N = N + 1 - \frac{1}{n} + o(n^{-1}). \quad (16.19)$$

Numerically, $\Phi(1/2) = 0.553771275888810\dots$

Corollary 16.4 (The lobe envelope is not two-sided). *For the sequence of lobe heights M_m ,*

$$0 < \limsup_{m \rightarrow \infty} \frac{M_m}{E_{\kappa_\infty}(m)} < \infty, \quad \liminf_{m \rightarrow \infty} \frac{M_m}{E_{\kappa_\infty}(m)} = 0. \quad (16.20)$$

Proof. The finite upper limsup follows from the global envelope. For the positive lower limsup, let $x_n = 2^n(2/3)$ and $m_n = \lfloor x_n \rfloor$. The maximum of the lobe containing x_n is at least $|\Phi(x_n)|$, and (13.9) gives a positive normalized value. Replacing x_n by the nearby integer m_n changes E_{κ_∞} by a factor tending to one. For the liminf, use any fixed-offset subsequence $m = N + r$. The logarithm in (16.12) has twice the negative quadratic coefficient of $\log E_{\kappa_\infty}(N)$, so the quotient tends to zero. $\square$

16.3 Intermediate boundary scales

The boundary identity also interpolates between the deep fixed-offset valleys and the generic shell scale. Suppose $N = 2^{n-1}$, $z_n = o(N)$, and $z_n \rightarrow \infty$. If the tail quotient in (16.1) stays bounded above and below by positive constants, then

$$\log |\Phi(N + z_n)| = \log |\Phi(z_n)| + n \log \frac{z_n}{N} + o(n). \quad (16.21)$$

Assume further that $\log z_n \sim \alpha \log N$, with $0 \leq \alpha < 1$, and that z_n follows the typical log-square law. Substituting that law into (16.21) gives

$$\log |\Phi(N + z_n)| = -\frac{1 - \alpha + \alpha^2/2}{a} (\log N)^2 + O(\log N) + o((\log N)^2). \quad (16.22)$$

The coefficient decreases continuously from $1/a$ at $\alpha = 0$ to $1/(2a)$ as $\alpha \uparrow 1$. This conditional calculation explains why plots of lobe heights show a broad continuum rather than only two sharply separated bands.

17 Integer geometric ratios

The dyadic transform is one member of a natural family. For a real number $b > 1$, define

$$\Phi_b(z) := \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{b^n}\right). \quad (17.1)$$

The same normal-convergence argument as in Theorem 2.1 shows that Φ_b is entire. When b is an integer, the multiplication map $T_b x = bx \pmod{1}$ replaces doubling, and most of the dyadic architecture survives with explicit new constants.

17.1 Global q -Pochhammer product and zero extraction

Proposition 17.1 (Global spectral Pochhammer factorization). *For every real $b > 1$ and every $z \in \mathbb{C}$,*

$$\Phi_b(z) = \prod_{\ell=1}^{\infty} (z^2/\ell^2; b^{-2})_{\infty}. \quad (17.2)$$

More generally, the tail

$$R_{b,m}(z) := \prod_{n=m}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{b^n}\right)$$

satisfies

$$R_{b,m}(z) = \prod_{\ell=1}^{\infty} (z^2/(\ell^2 b^{2m}); b^{-2})_{\infty}. \quad (17.3)$$

Both products converge locally uniformly.

Proof. Euler's sine product gives

$$\operatorname{sinc}\left(\frac{\pi z}{b^n}\right) = \prod_{\ell=1}^{\infty} \left(1 - \frac{z^2}{\ell^2 b^{2n}}\right).$$

On each compact set,

$$\sum_{n=0}^{\infty} \sum_{\ell=1}^{\infty} \frac{|z|^2}{\ell^2 b^{2n}} \leq C \frac{\zeta(2)}{1 - b^{-2}} < \infty.$$

Thus the double product is normally convergent and can be rearranged, including at points where one of its factors vanishes. Grouping first by ℓ gives (17.2). Replacing z by z/b^m gives (17.3). $\square$

For $J \geq 1$, let

$$H_J^{(s)} = \sum_{\ell=1}^J \ell^{-s}, \quad d_{J,k} = \frac{\zeta(2k) - H_J^{(2k)}}{k}.$$

Theorem 17.2 (Arbitrary-radius zero extraction). *Let $b > 1$, $m \geq 0$, and $J \geq 1$ be integers. If $|z| < (J+1)b^m$, then*

$$\begin{aligned} R_{b,m}(z) &= \prod_{\ell=1}^J (z^2/(\ell^2 b^{2m}); b^{-2})_{\infty} \\ &\quad \times \exp \left[- \sum_{k=1}^{\infty} \frac{d_{J,k}}{1 - b^{-2k}} \left(\frac{z}{b^m} \right)^{2k} \right]. \end{aligned} \quad (17.4)$$

The power series in the exponent has radius exactly $(J+1)b^m$.

Proof. Separate the factors $1 \leq \ell \leq J$ in (17.3). Put $u = z/b^m$. For $|u| < J+1$, the remaining factors satisfy $|u|^2/(\ell^2 b^{2n}) < 1$, and absolute convergence permits expansion and interchange of all sums:

$$\begin{aligned} &\log \prod_{\ell=J+1}^{\infty} \prod_{n=0}^{\infty} \left(1 - \frac{u^2}{\ell^2 b^{2n}} \right) \\ &= - \sum_{k=1}^{\infty} \frac{u^{2k}}{k} \left(\sum_{\ell=J+1}^{\infty} \ell^{-2k} \right) \left(\sum_{n=0}^{\infty} b^{-2nk} \right) \\ &= - \sum_{k=1}^{\infty} \frac{d_{J,k}}{1 - b^{-2k}} u^{2k}. \end{aligned}$$

Exponentiation proves (17.4). Finally,

$$\zeta(2k) - H_J^{(2k)} = (J+1)^{-2k} \left(1 + O \left(\left(\frac{J+1}{J+2} \right)^{2k} \right) \right).$$

The root test therefore gives radius $J+1$ in u , equivalently $(J+1)b^m$ in z . Analytically, the boundary is forced by the first unextracted zeros at $u = \pm(J+1)$. $\square$

The corrected dyadic Gamma–Pochhammer tail in Theorem 2.6 is the case $b = 2$, $J = 1$. The two indices m and J provide independent numerical controls: move farther into the small-argument tail, or extract more zero layers explicitly.

17.2 Zeros for an integer ratio

Assume now that $b \geq 2$ is an integer. For $M \neq 0$, define

$$\nu_b^{\text{pow}}(M) = \max\{j \geq 0 : b^j \mid M\}. \quad (17.5)$$

For prime b this is the usual b -adic valuation; for composite b it records whole powers of b , which is exactly what the product requires.

Corollary 17.3 (Integer-ratio zero divisor). *The nonzero zeros of Φ_b are precisely the nonzero integers. At $M \in \mathbb{Z} \setminus \{0\}$, the multiplicity is*

$$1 + \nu_b^{\text{pow}}(|M|). \quad (17.6)$$

Proof. The first factor $\text{sinc}(\pi z)$ vanishes at every nonzero integer and nowhere else on the real axis; all later zero sets are subsets of the integers. A level n contributes a zero at M exactly when $b^n \mid M$. The admissible levels are $0 \leq n \leq \nu_b^{\text{pow}}(|M|)$, proving the multiplicity. $\square$

17.3 Exact shell identity and fixed rays

Let

$$\phi(t) = \log |\sin(\pi t)|, \quad T_b(t) = bt \pmod{1}. \quad (17.7)$$

Theorem 17.4 (Exact b -adic shell identity). *Let $b \geq 2$ be an integer, $N \geq 0$, and $1 \leq y < b$. Then*

$$\begin{aligned} \log |\Phi_b(b^N y)| &= \log |\Phi_b(y)| - N \log(\pi y) - \frac{\log b}{2} N(N+1) \\ &\quad + \sum_{k=1}^N \phi(T_b^k \{y\}), \end{aligned} \quad (17.8)$$

where $\{y\}$ is the fractional part. At a zero, both sides are understood as $-\infty$.

Proof. Reindex the product:

$$\Phi_b(b^N y) = \Phi_b(y) \prod_{k=1}^N \text{sinc}(\pi b^k y). \quad (17.9)$$

Because b is an integer, $|\sin(\pi b^k y)| = |\sin(\pi T_b^k \{y\})|$. Taking absolute values and logarithms, and using $\sum_{k=1}^N k = N(N+1)/2$, gives (17.8). $\square$

Theorem 17.5 (Fixed-ray and typical decay for integer b). *Let $1 \leq y < b$, suppose $\Phi_b(y) \neq 0$, and assume that the finite limit*

$$\alpha_b(y) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N \phi(T_b^k \{y\}) \quad (17.10)$$

exists. For $x_N = b^N y$,

$$\log |\Phi_b(x_N)| = -\frac{(\log x_N)^2}{2 \log b} - \kappa_b(\alpha_b(y)) \log x_N + o(\log x_N), \quad (17.11)$$

where

$$\kappa_b(\alpha) = \frac{\log \pi + \frac{1}{2} \log b - \alpha}{\log b}. \quad (17.12)$$

For Lebesgue-almost every $y \in (1, b)$,

$$\boxed{\kappa_{\text{typ}}(b) = \frac{1}{2} + \log_b(2\pi)}. \quad (17.13)$$

Proof. Put $B = \log b$. By (17.8) and the assumed Birkhoff average,

$$\log |\Phi_b(b^N y)| = -\frac{B}{2} N^2 + N \left(\alpha_b(y) - \log(\pi y) - \frac{B}{2} \right) + o(N) + O(1).$$

Since $N = (\log x_N - \log y)/B$, substitute this expression and expand. The mixed terms containing $(\log y)(\log x_N)$ cancel, leaving exactly (17.11) and (17.12).

The map T_b preserves Lebesgue measure and is ergodic. Since $\phi \in L^1(0, 1)$ and

$$\int_0^1 \phi(t) dt = -\log 2,$$

Birkhoff's theorem gives $\alpha_b(y) = -\log 2$ for almost every y . Inserting this value in (17.12) yields

$$\kappa_b(-\log 2) = \frac{1}{2} + \frac{\log(2\pi)}{\log b},$$

which is (17.13). At $b = 2$, it reduces to $3/2 + \log_2 \pi = \kappa_0$. $\square$

17.4 Exact covariance and the general fluctuation constant

Put

$$\psi(t) = \log |2 \sin(\pi t)|, \quad S_N^{(b)}(t) = \sum_{j=0}^{N-1} \psi(T_b^j t). \quad (17.14)$$

As before, ψ has mean zero and Fourier series (6.3).

Theorem 17.6 (Exact integer-base variance). *For every integer $b \geq 2$ and every $j \geq 0$,*

$$\int_0^1 \psi(t) \psi(T_b^j t) dt = \frac{\pi^2}{12b^j}. \quad (17.15)$$

Consequently, for $N \geq 1$,

$$\begin{aligned} \int_0^1 (S_N^{(b)}(t))^2 dt &= \frac{\pi^2}{12} \left[N + 2 \sum_{j=1}^{N-1} (N-j)b^{-j} \right] \\ &= \frac{\pi^2}{12} \left[\frac{b+1}{b-1} N - \frac{2b}{(b-1)^2} + \frac{2b^{1-N}}{(b-1)^2} \right]. \end{aligned} \quad (17.16)$$

The asymptotic variance per level is therefore

$$\boxed{\sigma_b^2 = \frac{\pi^2 b + 1}{12 b - 1}}. \quad (17.17)$$

Proof. Use the L^2 Fourier series

$$\psi(t) = - \sum_{m=1}^{\infty} \frac{\cos(2\pi m t)}{m}.$$

After replacing t by $b^j t$, orthogonality leaves exactly the pairs of frequencies $m = b^j \ell$ and ℓ . Therefore

$$\int_0^1 \psi(t) \psi(T_b^j t) dt = \frac{1}{2} \sum_{\ell=1}^{\infty} \frac{1}{b^j \ell} \frac{1}{\ell} = \frac{\pi^2}{12b^j},$$

proving (17.15). Stationarity gives

$$\|S_N^{(b)}\|_2^2 = N C_0 + 2 \sum_{j=1}^{N-1} (N-j) C_j, \quad C_j = \frac{\pi^2}{12b^j}.$$

The finite arithmetic-geometric sum is

$$\sum_{j=1}^{N-1} (N-j)b^{-j} = \frac{N}{b-1} - \frac{b}{(b-1)^2} + \frac{b^{1-N}}{(b-1)^2}.$$

Substitution gives (17.16), and division by N followed by $N \rightarrow \infty$ gives (17.17). $\square$

The martingale structure is again explicit. Let

$$(\mathcal{P}_b f)(x) = \frac{1}{b} \sum_{r=0}^{b-1} f\left(\frac{x+r}{b}\right) \quad (17.18)$$

be the Lebesgue Perron operator. Its action on Fourier modes implies $\mathcal{P}_b \psi = \psi/b$. Define

$$h_b = \frac{\psi}{b-1}, \quad d_b = \psi + h_b - h_b \circ T_b = \frac{b\psi - \psi \circ T_b}{b-1}. \quad (17.19)$$

Then $\mathcal{P}_b d_b = 0$, and telescoping gives

$$S_N^{(b)} = \sum_{j=0}^{N-1} d_b \circ T_b^j - h_b + h_b \circ T_b^N. \quad (17.20)$$

The variables $d_b \circ T_b^j$ are orthogonal reverse martingale differences. Comparing (17.20) in L^2 with (17.16) shows

$$\|d_b\|_2^2 = \sigma_b^2. \quad (17.21)$$

Indeed, the coboundary has bounded L^2 -norm, while the martingale sum has squared norm $N \|d_b\|_2^2$; divide by N and pass to the limit.

Corollary 17.7 (Integer-base CLT and LIL). *For every integer $b \geq 2$,*

$$\frac{S_N^{(b)}}{\sigma_b \sqrt{N}} \Rightarrow N(0, 1). \quad (17.22)$$

For Lebesgue-almost every t ,

$$\limsup_{N \rightarrow \infty} \frac{S_N^{(b)}(t)}{\sqrt{2\sigma_b^2 N \log \log N}} = 1, \quad \liminf_{N \rightarrow \infty} \frac{S_N^{(b)}(t)}{\sqrt{2\sigma_b^2 N \log \log N}} = -1. \quad (17.23)$$

Proof. The logarithmic singularities of ψ , and hence of d_b , have exponential tails; therefore $d_b \in L^p$ for every finite p . The identity $\mathcal{P}_b d_b = 0$ makes $(d_b \circ T_b^j)$ a stationary ergodic reverse martingale-difference sequence. Reading each finite row in reverse order, exactly as in the proof of Theorem 6.4, turns it into an ordinary martingale array. Its conditional quadratic variation divided by N converges almost surely to $\|d_b\|_2^2 = \sigma_b^2$, and square integrability gives the conditional Lindeberg condition. Brown's theorem [4] therefore gives (17.22). Cuny's compact LIL for stationary ergodic reverse martingale differences [5] gives (17.23) directly.

Finally, the two terms $-h_b + h_b \circ T_b^N$ in (17.20) are negligible. Their exponential tails imply, by Borel–Cantelli,

$$h_b \circ T_b^N = o(\sqrt{N \log \log N}) \quad \text{almost surely,}$$

and they are $o(\sqrt{N})$ in probability. Thus the martingale limits transfer to $S_N^{(b)}$. $\square$

What the formula says

Two formulas summarize the integer-ratio universality class:

$$\kappa_{\text{typ}}(b) = \frac{1}{2} + \log_b(2\pi), \quad \sigma_b^2 = \frac{\pi^2}{12} \frac{b+1}{b-1}.$$

The first comes from the triangular product of b -adic denominators and the universal log-sine mean $-\log 2$. The second comes from exact Fourier-frequency matching under multiplication by b .

18 Conclusion: a decay dictionary rather than one decay rate

The Fourier transform of the up-function has one deterministic mechanism and several statistical mechanisms. The deterministic mechanism is the triangular dyadic denominator, which produces

$$\exp\left(-\frac{(\log x)^2}{2 \log 2}\right).$$

After this factor is removed, a lacunary sine product remains. Different ways of measuring that product produce different powers of x .

Question	numerator scale per level	correct power	main tool
Typical point / geometric mean	$1/2$	$\kappa_0 = \frac{3}{2} + \log_2 \pi$	log-sine integral and Birkhoff
Arithmetic shell mean	$\rho_1 = 0.6613226020 \dots$	$\kappa_1 = 2.7480700148 \dots$	weighted transfer operator
Root-mean-square	$1/\sqrt{2}$	$\kappa_2 = \log_2(2\pi)$	finite Parseval identity
Uniform pointwise envelope	$\sqrt{3}/2$	$\kappa_\infty = \frac{3}{2} + \log_2(\pi/\sqrt{3})$	sharp two-step sine inequality
Raw maximum on $[R, 2R]$	moving boundary layer	κ_∞ plus a second log-normal loss	transition coordinates and Lambert W
Fixed-offset lobe after 2^n	terminal integer zero	log-square coefficient $1/\log 2$	exact dyadic-boundary identity

Three qualifications are essential.

First, the power κ is not a minor logarithmic correction. The four principal powers differ by nonzero constants, so replacing one by another changes the logarithm of the amplitude by a term of order $\log x$. No factor $(\log x)^p$ can repair that error.

Second, ordinary Cesàro averaging has an endpoint phase. The natural fixed gauge is correct on complete dyadic shells and at dyadic endpoints, but a partial final shell leaves a nonconstant singular profile. Bounded-distortion corrections cannot remove it. A shell-adaptive decreasing analytic gauge can, because its increasingly fine microstructure tracks the finite-level mass distribution. If a fixed multiplicative geometry is preferred, the logarithmic mean is the canonical endpoint-independent average.

Third, local maxima are not asymptotically equivalent to a single envelope. The period-two ray proves the sharp upper scale, while the fixed-offset valley theorem produces lobe heights with twice the generic negative log-square coefficient. The sequence of normalized lobe heights therefore has positive limsup and zero liminf.

The integer-ratio family shows which parts of the theory are dyadic accidents and which are structural. The log-square coefficient becomes $1/(2 \log b)$, the typical power becomes $1/2 + \log_b(2\pi)$, and the exact fluctuation variance becomes $\pi^2(b+1)/(12(b-1))$. The same architecture—geometric denominator, expanding map, weighted product, transfer spectrum—persists.

Proof status

The elementary product algebra, zero structure, exact renormalizations, covariance sums, Parseval identity, Gelfond envelope, Lambert optimization, peak-area estimate, and valley laws are proved directly in this article. Four general tools are imported with their roles made explicit: Birkhoff's theorem, the Ruelle–Perron–Frobenius spectral-gap theorem, the martingale CLT/compact LIL, and Carleman–Höischen approximation. The arithmetic-mean Perron-root enclosure is an exact rational/Sturm certificate; longer decimal expansions and the finite- q plot are numerical evidence.

A Notation at a glance

Symbol	Meaning
$\Phi = \widehat{\text{up}}$	Fourier transform of Rvachev's up-function under $e^{-2\pi i t \xi}$
a	$\log 2$
T	doubling map $x \mapsto 2x \pmod{1}$
$s(x)$	$ \sin(\pi x) $
$B_n(y)$	the n -factor forward sine product appearing in the dyadic identity
$Q_n(x)$	$\prod_{j=0}^{n-1} s(T^j x)$
$E_\kappa(x)$	$x^{-\kappa} \exp(-(\log x)^2 / (2 \log 2))$
$W_\kappa(y)$	the bounded fixed-phase factor in the normalized master identity
$\mathcal{L}_q$	weighted inverse-branch operator in (7.2)
ρ_q	Perron root of $\mathcal{L}_q$
Λ_q	$\rho_q^{1/q}$, the L^q numerator growth per dyadic level
κ_q	$(\log \pi + a/2 - \log \Lambda_q)/a$
M_m	unique maximum of $ \Phi $ on $[m, m+1]$
$\mathfrak{M}_k$	maximum of $ \Phi $ on $[2^k, 2^{k+1}]$
$\mathcal{H}(y)$	$ \Phi(y/2) $, the nonvanishing tail in annular and valley formulas

B Exact Sturm certificate for the arithmetic-mean Perron root

This appendix makes the computer-assisted part of [Theorem 9.1](#) reproducible. Every coefficient is a rational integer, and SymPy's Sturm implementation performs exact Euclidean division over $\mathbb{Q}[t]$; no floating-point number is created.

For reference, after primitive integer normalization the three sign polynomials are

```

D(t) =
250000*t^12 - 376189*t^11 + 1223628*t^10 - 1376295*t^9
+ 3750000*t^8 - 9194*t^7 + 5552744*t^6 + 9194*t^5
+ 3750000*t^4 + 1376295*t^3 + 1223628*t^2 + 376189*t + 250000

P_lo(t) =
-23272139250000*t^12 + 49876177398523*t^11
-103087512595796*t^10 + 209100693840065*t^9
-239628013750000*t^8 + 229880069863558*t^7
-266170630808408*t^6 + 228664130136442*t^5
-109446038750000*t^4 + 27080706159935*t^3
-3087512595796*t^2 + 123822601477*t + 535750000

P_hi(t) =
23274160250000*t^12 - 49879218510399*t^11
+103097404404548*t^10 - 209111819808845*t^9
+239658328750000*t^8 - 229880144187854*t^7
+266215519190904*t^6 - 228664055812146*t^5
+109476353750000*t^4 - 27069580191155*t^3
+3097404404548*t^2 - 120781489601*t + 1485250000
    
```

They satisfy

$$D(t) > 0, \quad P_{\text{lo}}(t) > 0, \quad P_{\text{hi}}(t) > 0 \quad (0 \leq t \leq 1).$$

Since

$$P_{\text{lo}} \propto N - \frac{66126807}{10^8} D, \quad P_{\text{hi}} \propto \frac{66134891}{10^8} D - N,$$

with positive proportionality constants, these inequalities are exactly the claimed lower and upper bounds for $R = N/D$.

```

"""Exact certificate for Proposition B.1; requires SymPy."""
from __future__ import annotations

import sympy as sp

t = sp.symbols("t")
u = 2 * t / (1 + t**2)
v = (1 - t**2) / (1 + t**2)

def p(x: sp.Expr) -> sp.Expr:
    return (
        1
        + sp.Rational(376189, 10**6) * x
        - sp.Rational(69093, 10**6) * x**2
        + sp.Rational(15483, 10**6) * x**3
    )

R = sp.cancel((u * p(u) + v * p(v)) / (2 * p(2 * u * v)))
num, den = sp.fraction(R)
lo = sp.Rational(66126807, 10**8)
hi = sp.Rational(66134891, 10**8)

def primitive_integer_polynomial(expr: sp.Expr) -> sp.Poly:
    q = sp.Poly(expr, t, domain=sp.QQ)
    common_denominator = sp.ilcm(*[c.q for c in q.all_coeffs()])
    integer_q = sp.Poly(
        sp.expand(q.as_expr() * common_denominator), t, domain=sp.ZZ
    )
    return integer_q.primitive()[1]

D = primitive_integer_polynomial(den)
P_lo = primitive_integer_polynomial(num - lo * den)
P_hi = primitive_integer_polynomial(hi * den - num)

def sign_variations(chain: list[sp.Expr], x: int) -> int:
    signs: list[int] = []
    for q in chain:
        value = q.subs(t, x)
        if value > 0:
            signs.append(1)
        elif value < 0:
            signs.append(-1)
    return sum(a != b for a, b in zip(signs, signs[1:]))

for name, q in (("D", D), ("P_lo", P_lo), ("P_hi", P_hi)):
    chain = sp.sturm(q.as_expr(), t)
    V0 = sign_variations(chain, 0)
    V1 = sign_variations(chain, 1)
    assert q.eval(0) > 0
    assert V0 == V1 == 6
    assert q.count_roots(0, 1) == 0
    print(name, "degree", q.degree(), "V(0),V(1)=", V0, V1,
          "endpoint values=", q.eval(0), q.eval(1))

```

The exact output is

```
D degree 12 V(0),V(1)= 6 6 endpoint values= 250000 16000000
P_lo degree 12 V(0),V(1)= 6 6 endpoint values= 535750000 34288000000
P_hi degree 12 V(0),V(1)= 6 6 endpoint values= 1485250000 95056000000
```

Sturm's theorem says that $V(0) - V(1)$ is the number of distinct roots in $(0, 1)$. Thus each polynomial is root-free there; its positive value at zero fixes its sign throughout the closed interval.

C How the numerical spectrum was computed

The decimals in the finite- q table are orientation data, not interval certificates. A stable way to reproduce them is Chebyshev collocation.

Choose the shifted Chebyshev–Lobatto nodes

$$x_j = \frac{1}{2} \left(1 - \cos \frac{j\pi}{N} \right), \quad 0 \leq j \leq N. \quad (\text{C.1})$$

Let $I_N(z)$ be the row vector of barycentric interpolation weights that evaluates the unique degree- N interpolant at z . The matrix representing $\mathcal{L}_q$ on nodal values is

$$L_{q,N}[j, :] = \frac{1}{2} \sin^q \left(\frac{\pi x_j}{2} \right) I_N(x_j/2) + \frac{1}{2} \cos^q \left(\frac{\pi x_j}{2} \right) I_N((x_j + 1)/2). \quad (\text{C.2})$$

Its Perron eigenvalue converges rapidly to ρ_q . The practical checks used here were:

1. repeat with increasing even degrees N , retaining only digits stable across several successive degrees;
2. verify positivity and symmetry of the leading eigenvector;
3. check $q = 2$ against the exact eigenpair $\rho_2 = 1/2$, $h_2 = 1$;
4. check monotonicity of $\rho_q^{1/q}$ and the endpoint trend toward $\sqrt{3}/2$;
5. for $q = 1$, verify that the numerical root lies inside the exact Sturm and published enclosures.

These tests detect ordinary truncation or transcription errors, but they do not turn a floating-point eigenvalue into a rigorous interval. Rigorous values can instead be obtained from positive polynomial sub- and supersolutions and Collatz–Wielandt bounds, as in [Theorem 9.1](#).

D Proof-dependency ledger

Result family	Status	External input or exact certificate
Entire product, alternative products, zeros, local coefficients	self-contained	normal convergence, Euler and Viète products
Unique lobe maximum, sign law, peak–area estimate	self-contained	termwise logarithmic differentiation justified locally
Dyadic factorization and Thue–Morse bridge	self-contained	finite reindexing and binary polynomial identity
Geometric shell mean	self-contained	exact log-sine integral
Almost-everywhere typical exponent	imported theorem	Birkhoff ergodic theorem for doubling
Exact covariance and martingale decomposition	self-contained	Fourier orthogonality and beta-integral differentiation

Result family	Status	External input or exact certificate
CLT and LIL	imported probability	Brown martingale CLT and Cuny compact LIL
Finite- q shell limits	imported functional analysis	Ruelle–Perron–Frobenius spectral gap for $\mathcal{L}_q$
Root-mean-square numerator rate	self-contained	finite Parseval identity
Arithmetic Perron-root bracket	exact computer algebra	rational Collatz–Wielandt inequality and Sturm certificate in Section B
Cesàro endpoint profile and bounded-distortion obstruction	spectral plus elementary	RPF rank-one limit, weak convergence, elementary measure estimates
Adaptive smooth gauge	self-contained construction	finite shell flattening and monotonicity estimates
Analytic adaptive gauge	imported approximation	Carleman–Hoischen simultaneous approximation of a function and derivative
Sharp uniform envelope	self-contained	elementary sharp sine inequality
Annular Lambert law	self-contained	exact transition coordinates, dense preimages, one-variable optimization
Odd-multiple and fixed-offset valleys	self-contained	exact boundary factorizations and uniform Laplace principle
Integer-base fluctuation laws	algebra self-contained; limits imported	exact covariance plus the same martingale limit theorems

The starting source and its auxiliary verification materials are maintained in the ProveIt repository [10]. This rewritten version deliberately omits the chronology of earlier reports from the main exposition: provenance is useful for an audit, but it obscures the mathematical dependency graph for a reader learning the result.

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